Opening AI the Data-centric Way:

The Case for Data Institutions in Democratising the Design & Development of ML Datasets

Henry Baker, MPP

2024

Supervisor: Prof. Dr. Joanna Bryson

Word count: 12,941

Acknowledgements

To my supervisor Prof. Dr. Joanna Bryson, to my colleagues in the ODI's Data-centric AI project, to Dad.

Contents

Abbreviations

AI	Artificial Intelligence
ML	Machine Learning
DI	Data Institution
FOSS	Free & Open Source Software
IP	Intellectual Property
ToS	Terms of Service

Executive Summary

This thesis examines the role of dedicated Data Institutions (DIs) in the design and development of machine learning (ML) datasets, emphasising their potential to 'democratise AI'. The analysis critiques the current *open AI* approach and its limitations, proposing that as an institutionalised form of commons-based data governance, DIs offer a structured democratic approach to data-centric AI policy-making.

As an exploratory first pass, the research draws on multiple literatures, including critical political economy, data management, and governance theorising, to provide a normative and theoretical evaluation of DIs as an institutional template. This thesis proposes that integrating DIs into the ML value chain would leverage the strategic significance of datasets to create public value and insert a decentralised, community-accountable governance layer into the otherwise opaque ML development process. It sets out an agenda for DIs' collective governance mechanisms and fiduciary responsibilities to improve current (mal)practice in dataset development around data quality, usage, sustainability, and embedding of good governance practices.

Chapter 1

Introduction

Open AI's FOSS fallacy

This research developed as a response to perceived limitations in the debates happening around open source and its relationship to AI systems (hereafter *'open AI'*). Specifically it came from the premise that there are more effective, more responsible, and less publicised policy approaches to achieve the fundamental objectives motivating the 'open' vs 'closed' AI debate, than are currently duly considered. Accordingly, this thesis chooses to (re)focus on the *democratisation* of AI. I operationalise 'democratisation of AI' as the establishment of legitimate mechanisms along the value chain for supporting equitable, inclusive and sustainable distributions of benefit and access around AI development, use of AI, value-generated by AI, and governance of AI systems themselves. This account of AI's democratisation starts from the belief that these outcomes can be at least partially achieved at an upstream technology-governance level. Moving beyond the constraining terms of the *open AI* debate, I consider Data Institutions (DIs) from a data-centric AI governance perspective, proposing DIs as a promising candidate for institutionalising democratic principles into the governance of AI technologies and resources.

As a nascent field of policy study, focusing on societal-level outcome objectives for AI governance remains fruitful. Given the path dependency coalescing around *open AI* and AI risk debates, now is an appropriate time to widen the lens of policy possibilities as we collectively grapple with the implications of AI technologies for economic growth, redistribution, social development, representation and meaning. (Admittedly, the trade-off for the ambitious breadth of this analysis will be its limited depth). In this spirit of critical engagement, I propose that *democratising*

AI builds on three selected limitations within the *open AI* debate.

From a political-theory perspective, AI democratisation is not solely about the free deployment of AI without regard for social consequence. Rather, democratisation entails collective decision-making power over how AI is to be developed and deployed. The 'narrow' democratisation of *open AI*, conflicts with these 'broader' democratic ideals, as unstructured access to AI systems could hinder societies from restricting those uses they deem undesirable (Chan et al., 2023).

From a political-economy perspective, the liberatory potential of *open AI* is bounded by the reality of near-term resource constraints (compute, data, and to a lesser extent human capital). The ability of platform incumbents operating across parallel digital markets to cross-leverage scale and data, risks the instrumentation and capture of *open AI* discourses by entrenched actors. For excellent descriptive analyses of strategic 'open washing' along ML value chains see both Gray-Widder et al. (2023) and Liesenfeld et al. (2023); also L. M. Khan (2017) for an antitrust rebuke to integrating across data markets.

Finally, from a framing perspective, *open AI* debates are *a priori* rooted in a framework for evaluating the extent and value of openness by means of a risk-benefit calculus over downstream use cases. Democratisation as an evaluative criteria moves beyond this FOSS-derived mode of thinking. As an analytical lens, risk diverts attention towards distributive policy choices at the system-output level, and 'naturally' suggests restrictive point-of-service regulation. Accordingly, policy-makers are currently legislating model capabilities (Bommasani et al., 2022), post-release conformity (Mökander et al., 2022), risk categorisation (EU AI Act), and access regimes (Bostrom, 2017; Shevlane, 2022). This is because risk-calculus thinking encourages binary-choice conceptualisations of AI's *problem of openness*. Recent *open AI* thinking has begun to introduce more graduated instruments of governance - including gradient-release processes (DPGA, 2023; Solaiman, 2023), risk categorisation (EU AI Act), and divers ToS models (Gorwa and Veale, 2024) - yet it remains the case that viewing openness primarily through the lens of access and associated risks lacks analytical purchase.

I propose that integrative, upstream policy interventions also have a role in AI governance. It is only after disaggregating AI systems and evaluating each component by stakeholders affected and values encoded, that requisite policy nuance is to be found. Whereas compute, algorithms, and weights remain access issues (Liesenfeld et al., 2023), I propose that data governance - *how* data is generated, collected, licensed, processed, owned, stewarded, augmented, reproduced, and distributed -

presents a variety of decision dimensions and policy positions for democratic considerations.

Pure-FOSS modes of AI governance over models as end-products is suboptimal. It conceptualises blunt, technically-oriented policy instruments for intractable, social problems. In its place, pursuing democratic accountability across the value chain allows for the necessary nuance and integrative thinking for successful AI policy-making.

The logic of data institutions

As a developing and much-contested area of digital policy, there are a range of overlapping and often-competing naming conventions and governance claims made by proponents of DIs. *Data Trusts* were an early proposal, and one still put forward by the Data Trusts Initiative, *Data Collaboratives* are championed by The GovLab, more market-oriented *Data Intermediaries* are preferred by the UK government in its 2020 National Data Strategy and in reports from the Centre for Data Ethics and Innovation (2019), *Data Spaces* and *Data Altruism Organisations* represent the EU Commission's dual proposals, while the Oxford Margin Programme deploys the pithy *Ethical Web and Data Architectures*. This plurality of naming devices ultimately reflects gradations of legal distinctions and policy emphases within the concept of DIs writ large. This diversity matters at the implementation level, both in terms of legal justification (see O'Hara (2020) and Olorunju and Adams (2023)) and domain-fit design (see Delacroix and Lawrence (2019) and GPAI (2023b)), but it has little to say with regard to DIs' common institutional logic, whereby data-generating individuals and communities delegate fiduciary responsibility for stewarding their data to a given management function, from which DIs then engage with data users (here: ML developers) on the community's behalf (Keller, 2021).

At this thesis' level of abstraction, the primary significance of the many DI types is to signify their collective flexibility as a core institutional template for meeting a variety of context-dependent requirements. Below the apparent institutional variety, the dual concerns common to all DI types are *delegated fiduciary responsibility* and *collective or commons-based governance* mechanisms (Keller, 2021). DIs offer policy-makers the ability to build on top of this core institutional template to incorporate a wide range of policy and technical functions. This flexibility means DIs can be deployed to meet divers data governance demands across divergent local conditions. Since I restrict this analysis to high-level normative concerns, I treat these institutional variations as parallel instances of *bottom-up Data Institutions* - a deliberately catch-all term proposed by the ODI (Hardinges and Keller, 2021b) drawing on Delacroix and Lawrence (2019), and recently adopted by GPAI (2023b).[1] As such, DIs' primary institutional logic in commons-based data governance theorising is to structure a decentralised community-accountable layer of decision-making over resource allocation (Delacroix et al., 2020). This research applies that core institutional logic to ML datasets as an AI resource.

Data institutions & democratisation of ML datasets

This research seeks to widen the scope of possible intervention points into AI governance. For this, ML datasets and their collection present a strategic policy surface within otherwise opaque ML value chains - and DIs present an ideal template for their governance.

I propose that ML-relevant use cases for DIs extend beyond their well-theorised ability to redistribute value across imbalanced digital markets. My proposal is that establishing dedicated DIs for the design, development and stewarding of ML datasets inserts a new (decentralised, community-accountable) governance layer. This new composite 'layer' offers *two-fold* public value creation: alongside integrating divers voices into foundational value-encoding and direction-setting processes at the heart of AI development, dataset DIs would also serve as institutional platforms for a range of contextually-appropriate downstream service offerings and functionality - of benefit to both data producers and users. This might include improved quality-assurance, semantic interoperability, responsible management of consent and licensing regimes, or financial sustainability. This dual-sided utility would help policy-makers implementing DIs to gain broad-based stakeholder buy-in to successfully navigate the policy process, and ensure that taken collectively, this 'DI layer' could provide the necessary overhead and resources to support a data commons capable of sustaining future AI innovations.

A note on research design

As recent topics of policy study, collective data governance and ML value chains are slippery surfaces on which to carry out robust empirical analysis. Instead, this is

[1] In fact, there *are* DI types that remain outside of this definition. These DIs that centre individuals exercising active autonomy over their data are functionally separate, mostly marginal and not the concern of this analysis. I generically refer to 'DIs' to denote the majority of commons-based delegated decision-making institutions, except when context requires explicit identification otherwise. For alternative taxonomies, see Soni (2020) and Hardinges and Keller (2021a)).

intended as a first pass for further research to build upon. Rather than (reasonably) rely on heuristics comparing AI to nuclear weapons (Afina and Lewis, 2023) or other dual-use technologies (Ulnicane et al., 2021), or forecasting extrapolations through scaling laws (Kaplan et al., 2020; Villalobos et al., 2022) and risk analyses (Bostrom, 2017), I have chosen to undertake a comprehensive landscape review.

To make the case for commons-based DIs for ML datasets, I deploy multiple literatures - including critical political-economy analyses (CPE) of both digital economies and data as an economic-informational resource, practitioner-oriented data management and human-computer interaction (HCI) studies, as well as institutional theories of global governance (and its many critiques). Placing these literatures in conversation with one another allows knowledge produced in one to be deployed towards open questions raised in another. Just as early proponents of DIs linked critiques of data capitalism to the legal logic of a trust, the workhorse of this research is to 'match' critiques that have emerged across parallel research programmes, to the data governance possibilities afforded by institutionalised fiduciary responsibility. Specifically, I propose that the requirements for (and identified failings of) 'quality' ML datasets expressed in the data management and quantitative fairness literatures, inflected with critiques of current AI governance, can extend current theorising on collective data governance by suggesting technical functionality and policy principles for DI design to incorporate.

Search strategy

I first identified open Data-centric AI (DCAI) governance topics through reading publications of relevant organisations, journals, and research colloquia. I then conducted a systematic search of Google Scholar and selected outlets. This included the Data-centric Machine Learning Research (DMLR), Journal of Machine Learning Research (JMLR), Association for Computing Machinery (ACM), International Conference on Machine Learning (ICML), Nature Machine Intelligence, International Journal of Information Management (IJIM), Social Science Research Network (SSRN) - specifically the Computer Science Research Network (CompSciRN), Information Science Research Network (InfoSciRN), and Political Science Network (PSN). Papers were selected based on the relevance of their abstracts, before snowballing through selected references.

Constraints

This research is not intended as the final word on DIs and ML datasets. It does not make empirical claims over the precise nature of the relationship between commons-based data governance and ML development practice. Rather it flags the under-appreciated synergy between institutional theory and challenges faced in ML dataset curation.

This research avoids highly applied questions of DI design and implementation (such as sustainable funding arrangements, or technical/organisational architectures) which will vary according to the contextual logic of the 'task' being addressed. These are solved problems - if not at scale. For proofs-of-concept see Curry et al. (2022), "Data trusts Initiative" (n.d.), GPAI (2023b), and Tarkowski and Warso (2024) - among others.

Roadmap

I first consider datasets as strategic policy objects, before outlining the fields of study that investigate them. This provides an overview of the harms generated across the ML dataset lifecycle. Next, I give an account of past-and-current institutional theorising around DIs as a mode of commons-based data governance. The core proposition behind my research is to bring these two topics (ML datasets and their harms; DIs and their opportunities) into conversation with one another. Doing so makes the case for establishing DIs to steward ML datasets as a move to democratise AI across four interrelated dimensions of public value creation. Another way to conceptualise this is that collective data governance moves to close the *paradox of openness*, which remains unaddressed by the terms of the existing *open AI* debate. The four dimensions are as follows: DIs supply higher quality ML *data* and higher quality data *usage* in ML systems, DIs also structure sustainable incentives for perpetuating a digital commons and the AI systems trained on them, DIs' grassroots accountability and the distributed leverage they serve data-generating communities with promotes their inclusion and equity. Finally, I derive nine high-level principles for practitioners.

Chapter 2

Datasets & Their Study

Datasets as strategic policy objects

By unpacking the implications of literature exploring datasets as both constructed and strategic objects within ML value chains, I propose that establishing DIs to supply ML datasets would insert upstream, integrative policy surfaces into what are predominantly closed and opaque ML production pipelines.

At present, AI governance focuses on harms and risks at the point of inference. Efforts have been made to delineate (in)appropriate deployment use cases for predictive and algorithmic decision-making (Aragon et al., 2022; Birhane, 2022; Lum and Isaac, 2016), to gatekeep access (Gorwa and Veale, 2024; Shevlane, 2022; Shrestha et al., 2023), to mandate model requirements (Bommasani et al., 2024), and to implement algorithmic interventions altering system results towards unbiased outcomes (Bellamy et al., 2018; Dwork et al., 2012; Robertson, 2024; Wang et al., 2022; Zhao et al., 2018). This inference-oriented approach deemphasises the role of inputs and their governance on AI outcomes. However, recognition that "there is no AI without data" (Gröger, 2021; Snaith, 2023) as a recent point of departure for the data-centric AI approach, broadens the scope of policy interventions beyond just system outputs, towards encompassing datasets. As AI progress continues to be driven by Deep Learning architectures learning the underlying structure of a dataset *purely inductively*, Garbage In, Garbage Out (GIGO) - axiomatic of computer science since the 1960s - is now receiving due consideration with regards to data as representation for AI systems.

Data governance and AI governance represent interwoven realms - the former is prior and foundational to the latter (Verhulst et al., 2023). It follows then that the practice of dataset curation affords targeted intervention points into AI systems. Specifically, data enters AI systems through *datasets*, and the data management literature's focus on datasets over abstracted data, highlights their artificial, engineered and labour-intensive production processes. This insight suggests dataset curation's ability to motivate policy efforts around societal issues. Data availability constrains modelling effort towards tasks on which inference can be run (Leavy et al., 2020) to the extent that supporting dataset development around a 'task' or social issue supports technical efforts to address it. Further, that datasets are constructed, implies they can be strategically engineered towards socially-desirable outcomes (Oala et al., 2023; Verdegem, 2022). Considering this together with a framing of datasets as infrastructural to AI systems (themselves infrastructural to an 'AI economy' (M. Khan and Hanna, 2022)) further implies that dataset governance ought to incorporate broad societal engagement and active public scrutiny, based on economic orthodoxy applied to public infrastructure projects elsewhere (P. J. Singh, 2019).

The data management literature employs a dual taxonomy for ML datasets, which I use to articulate divergent governance approaches. *Training datasets* are used to teach models what they know - the majority are privately held, which serves as a barrier to entry for new market actors (Birkinbine, 2018; Spiekermann et al., 2021), and an obstacle to standardising audit practices (Raji et al., 2020; Shrestha et al., 2023). Accordingly, governance priorities relate to distribution and access, as well as content and harms contained therein. *Benchmark datasets* are curated and stewarded by 'task communities', and used to evaluate how well a model has learned to perform a technical 'task' in a proxy-population setting (Paullada et al., 2021), they are mostly openly hosted on sites such as Kaggle and MLCommons, or integrated into development environments such as TensorFlow or PyTorch. Koch et al. (2021) and Oala et al. (2023) highlight respectively the role of benchmark datasets in comparing between model types, and as estimation devices for how - and towards which ends - ML progress is being made. These task-oriented datasets connect policy challenges to fundamental ML research by providing the 'right datasets for the right problems' - or don't, leading to policy failure. Paullada et al. (2021) problematise how the development and task-focus of these strategic assets are driven by elite institutions, not always in alignment with the public interest. Governance priorities accordingly revolve around the incentives they structure within the ML research field.

The next generation of datasets tailored towards ML-specific purposes, open up new dimensions of quality across which DIs might create value. Efforts have been made to catalogue the technical requirements of 'AI-ready' datasets, characterised by dense metadata annotation (Gebru et al., 2021; Han et al., 2023), collaborative refinement, user preference and human feedback (Kirstain et al., 2023), and evolution over time (Schuhmann et al., 2022). Research by Priestley et al. (2023) into the burdensome process of developing safe, responsible 'AI-ready' datasets, and the extent to which those requirements are not currently met by profit-oriented ML developers, raise unanswered policy concerns that could inform the use and design of DIs as nodes in the ML value chain. Data in this developing framework is elevated from mere input resource - external to the model - to something like software code, iteratively evolving in conversation with model development (Jakubik et al., 2024). By taking data work seriously and reconceptualising ML datasets as dynamic, labour-intensive artefacts, purposefully developed by stakeholders with agendas, reflective of and shaped by decisions made across their lifecycle, I hope to reveal previously unconsidered dimensions to DIs' value proposition for the ML community and wider society.

Datasets in AI governance

Data as representation (and harms therein)

As an artefact of a decision to record a phenomena, data is biased to some irreducible degree. Nevertheless, as AI's salience as a public issue develops, there is growing consensus around the need to consider its ramifications and pursue greater mitigation efforts. The academic study of bias (here operationalised as *disparate results returned by sociotechnical systems across protected classes*) has spawned a robust quantitative fairness literature. Applied to the output of AI systems, this ranges from discussions of the use of ML in 'fair affirmative action' (Dwork et al., 2012), to gender bias in commercial deployment of classification systems (Buolamwini and Gebru, 2018), and whose values are considered ground truths (Crawford and Paglen, 2021). More recent *data-centric* efforts have considered best practice in ML data collection (Jo and Gebru, 2020), harms hidden inside training dataset content (Birhane et al., 2021), bias creep in both label encodings (Dixon et al., 2018; Ghai et al., 2020; Miceli et al., 2020) and outputs (Abid et al., 2021; Gehman et al., 2020; Wolfe and Caliskan, 2022), as well as appropriate legal accountability frameworks (M. Khan and Hanna, 2022). Similarly, various data-examination toolkits (Bellamy et al., 2018; Google, n.d.), validation approaches (Zhao et al., 2018), and benchmarks (Mazumder et al., 2023; Zhao et al., 2018) have been developed to evaluate datasets for biases. The salience of bias in AI is such that lively public debate around algorithmic decision-making in high-stakes public domains has sprung up, reaching into discussions of criminal justice and predictive policing (Lum and Isaac, 2016), hiring (Mann and O'Neil, 2016), critical infrastructure and finance (T. Lin, 2012), and the ethics and (lack of scientific validity) in face-to-sexuality prediction (Gehman et al., 2020; Wang et al., 2022), among other surveillance and inference-at-a-distance tasks. Nevertheless, it remains the case that *in practice*, ML practitioners and researchers are drawn from CompSci-affiliated engineering domains, and that funding incentives (public and private) are driven by national strategic and business imperatives. These incentives motivate demonstrable system improvements over social science rigour - with deleterious implications for bias. DIs offer a commercial-and-policy-viable alternative.

Most investigation of bias has thus far been directed at the system output level and the feedback loops engendered by algorithmic decision-making (see Abid et al. (2021), Buolamwini and Gebru (2018), Gehman et al. (2020), and Wolfe and Caliskan (2022)). This is a convenient location at which to take measurements, but it has led to neglecting how these harms were encoded into the data inputs in the first place. Despite constraints (most training datasets are proprietary, and their size make them intractable to thoroughly scrutinise (Bender et al., 2021)), a developing dataset auditing discipline is increasingly considering bias, class imbalance and other harms within these under-interrogated objects. This branch of investigation represents the most understood dimension of dataset-derived harms.

Recent (curtailed) efforts to audit ML training datasets reveal a myriad of representational harms (see Koch et al. (2021) for a comprehensive overview). Considering only the most commonly used training datasets Buolamwini and Gebru (2018) and Raji and Fried (2021) independently demonstrate under-representation of darker-skinned subjects in over 100 facial analysis datasets, while Wilson et al. (2019) show predictive inequality in object-recognition systems - specifically skin tones of pedestrians - as does DeVries et al. (2019) for regionally-sourced objects. Similarly Crawford and Paglen (2021) uncover millions of offensive labels within the ImageNet dataset; Prabhu and Birhane (2020) demonstrate non-consensual and pornographic imagery throughout the TinyImages dataset; Zhao et al. (2018) identify under-representation of female pronouns in the OntoNotes dataset; Kreutzer et al. (2022) find male labels for images of both outdoor athletes (irrespective of identifiable gender) and where a person is too small for identification. At a higher level of abstraction, Prabhu articulates an "abattoir effect" and Jo and Gebru (2020) critique a culture of overly-permissive harnessing of massive datasets, with laissez-faire attitudes around their development. Alternatively, Paullada et al. (2021) identify routine, implicit invocation of particular, nonneutral viewpoints in the design of tasks and labelling heuristics, and point to pervasive undervaluation of data work.

In contrast to other data-centric disciplines, ML practitioners (in aggregate) have been critiqued for lacking intentionality over the data they ingest, which has been linked to downstream harms. As such, datasets present a site of strategic contestation ripe for policy intervention. Operating at the point of collection, curation, distribution (and some preprocessing tasks), DIs and community-generated ML datasets offer a means of including a plurality of viewpoints, transparency and quality assurance mechanisms into this under-populated policy area.

Benchmarks as direction-setting devices for ML research

Beyond their contents, Koch et al. (2021) assert that benchmark datasets (and associated metrics) *direct* scientific ML progress and act as coordination devices for researchers around shared problems. Direction-setting is datasets' second strategic dimension. When institutionalising benchmarks, task communities endorse datasets as meaningful abstractions of a problem domain, and researchers are encouraged to make model choices that maximise performance accordingly. Alignment between benchmark datasets and "real world" tasks is thus critical; how alignment is measured, communicated and responded to offers a foothold for AI governance efforts.

Concerningly, Koch et al. (2021) also highlight field-wise concentration as a reduced number of elite (corporate, government, and academic) institutions shape the ML research agenda. Against this dynamic, establishing dedicated community-supported DIs for the continual development and maintenance of divers benchmark datasets, would better incorporate the interests and inputs of data-generating communities and associated stakeholders related to the 'task' (technical or social) at hand. DIs able to make data available for beneficiary-approved tasks (and to withhold where not) would collectively form a decentralised, composite layer of context-appropriate democratic oversight procedures. As units within a system of commons-governance, a network of DIs could legitimately guide the appropriate structuring of ML research incentives, according to democratic principles.

Datasets as bottleneck for leverage

The third strategic dimension of datasets for targeted AI governance is the oldest form of leverage: scarcity. Empirical research into ML scaling laws, suggests that while data and compute are today's principal constraints, of the two, data will remain (see Kaplan et al. (2020) for detailed analysis). As such, data access, sovereignty, and producer protection are the bedrock of sustainable AI (Curry et al., 2022). Society ought to be attentive to the ways that dataset owners and curators can be held accountable. Asserting more collective forms of control over access to datasets presents a point of democratic leverage in an otherwise concentrated industry structure.

The practice of ML dataset development

It is not just ML datasets' contents that make them strategic points of entry for AI governance. Literature investigating (mal)practice in the process of dataset development identifies multiple sites of harm generation across both original data collection-generation, and 'passive' data scraping and mining.

This line of research serves as a meta-oriented parallel field of study to dataset auditing, focused on the generative processes behind the artefacts themselves. Insights generated here suggest these processes are amenable to the kinds of oversight improvement that DI governance offers. These insights could further inform policymakers designing and implementing DIs by providing an architecture of existing pain points towards which to direct efforts.

Associated harms

Licences

Most major ML training datasets contain unlicensed, publicly available, internet-scraped data considered 'all rights reserved' by default. Despite the inclusion of robots.txt files or media clauses, the systematic inclusion of non-consenting sites is well demonstrated (M. Khan and Hanna, 2022). Currently it is simply infeasible for ambitious commercial ML projects to train only on content for which open licences exist (Chan et al., 2023). Benchmark datasets too are often drawn from a range of different sources, each with a different configuration of copyrights and permissions, although the problem is less severe and solution more manageable. Legal advocates have defended the use of copyrighted material under fair use and transformation clauses (Sag, 2019). This remains controversial. Datasets curated through *active* data collection-generation bypass some of these liability risks, but have been proven to generate a range of other harms (see M. Khan and Hanna (2022)). The decision to obtain or abide by appropriate licensing regimes is an irreducible source of tension and in near-constant violation by ML data users.

DIs able to license entrusted data return value to data producers and subjects.

While the economics of data licensing writ large remain intractable, a diverse ecosystem of bottom-up DIs could open up new areas of sufficiently overlapping interests between producers (data-generating individuals and communities) and consumers (ML developers) to support greater data sharing in an informed and licence-compliant manner (Delacroix and Lawrence, 2019).

DIs could further choose to assume responsibility for ensuring their datasets are licence-compliant for a stated range of ML use cases - benefiting both parties equally; producers regain downstream data control, consumers gain compliance assurance. Data-stewarding DIs could draw on recent efforts to develop a gradation of experimental AI-oriented open data licences (Dumas, 2023), to better represent their constituencies with greater granularity and freedom of movement, opening up dimensions of public value creation previously precluded by blunt licensing instruments (an alternative view is that recent proliferation of 'open and responsible' AI licensing regimes erodes longstanding open source norms).

Documentation, consent & distribution

Deficient documentation regarding datasets' consented use cases, the purpose of the original collection, or the conditions under which it was collected, are perennial challenges to ML developers. Investigations into major ML datasets such as the *Common Crawl*, *Faces in the Wild*, or *ImageNet* reveal poor documentation and repurposing of personal data for uses that were not anticipated when consent for collection was given (M. Khan and Hanna, 2022). By way of an early example, Radin (2017) was one of the first to document subsequent transpositions of a retracted (PIDD) dataset, tracing the harms derived from reusing data collected to refine algorithms that had nothing to do with that original understanding.

Concerningly, most commercial models are released without publicly sharing training datasets, and a recent audit of 1800 datasets highlights diminishing efforts among AI developers to document datasets used (Longpre et al., 2023). This is indicative of the fact that the release and distribution of datasets is typically not seen as part of the ML pipeline, leading social science critics contend that the ML field has yet to develop the data management practices required for the sensitive human data it is increasingly deployed to run inference on (Harvey and LaPlace, 2021; Raji et al., 2020; Roh et al., 2019).

Beyond the research ethics, deficient documentation and compromised data traceability is detrimental to responsible downstream model development (Gudivada et al., 2017; Longpre et al., 2023). Poor data genealogy hinders both the evaluation of data points' impact (*leverage*) on systems' outputs, and the ability to subsequently retract that data. These unintended afterlives (see accounts of derivative LFW, MS-Celeb-1M, and DukeMTMC datasets in Koch et al. (2021)) suggest the need for targeted policy interventions around managing consent and retraction points earlier in the data cycle. DIs institutionalise such mechanisms - as potential agents of enforcement and representative single-points-of-contact around which to develop a predefined set of responsibilities for supporting recourse, and in generating the documentation and traceability programmes necessary to facilitate post-hoc remediation.

Market concentration

The nature of data (alongside compute and human resources) make ML development a concentrated and opaque industry. As an economic and informational resource, data's utility lies in aggregation, and its collection benefits from economies of scale and cross-market leverage (Cilauro, 2021; Coyle and Manley, 2021). Data-driven web-2.0 markets favour anti-Schumpeterian "winner-took-all" rentier incumbents conducting massive data collection across multiple markets on privatised platform infrastructure. (for CPE analyses, Couldry and Mejias (2020), Meadway (2020), and Spiekermann et al. (2021); for the new antitrust perspective, L. M. Khan, 2017). Data's informational-relational aspects (Viljoen, 2021) make for privacy concerns (Zuboff, 2019) and IP infringement (Metz et al., 2024) among other sensitivities - all induce growing levels of secrecy. In ML, the result is concentration and opacity in commercial data sourcing. This undermines the public's ability to exercise decision-making power at strategic moments along the value chain, and puts data users in potential legal jeopardy (Bommasani et al., 2022, 2024).

Unlike antitrust and regulatory action, DIs for ML datasets address market concentration by leveraging data as an 'AI bottleneck' (while also accounting for its relational properties - unlike antitrust regulation). This efficiently negates the need for post-hoc distributive redress for the symptoms of market failure. Similarly, despite the proliferation of AI ethics standards (see Jobin et al. (2019)), ethical guidelines are no substitute for addressing the structural factors underlying the concentration of AI resources. Commons-based DIs in contrast change the dynamics of power to offer *integrative* industry restructuring - benefiting and compensating the majority of stakeholders while reassigning value closer to the true social costs/benefits.

Undervalued data work

The final dimension of harm in ML dataset development lies in the production process itself. Prevailing ML data practises abstract away the human labour involved in dataset production (Paullada et al., 2021) andd wilfully ignore the subjective non-neutral judgements and labelling heuristics (biases) humans necessarily bring to the encoding table (Jarrahi et al., 2023).

For labelled data, developers purchase data products from vendors (Scale AI, Surge AI, Snorkel.ai etc) or directly engage gig crowdworkers (via platforms such as Amazon Mechanicl Turk (AMT)). Although semi-supervised labelling and automated annotation tools are growing (as problematised by Jarrahi et al. (2023)), surveys of ML development practice highlight that cheap abundant gig labour remains the norm (Priestley et al., 2023). Beyond the propensity for labour abuses under the platformisation of economic precarity (see Miceli et al. (2020)), this has deleterious implications for data quality. Scholars have demonstrated how AMT's hierarchy positions data work and annotation as menial, leading to low quality results. This includes mistranslated text and mislabeled linguistic content (Kreutzer et al., 2022; Miceli et al., 2020), divergences in judgements across (and within) annotator pools, (Dixon et al., 2018; Kreutzer et al., 2022), subjective bias (Ghai et al., 2020; Hube et al., 2019), annotation that mischaracterise the real-world tasks the dataset supposedly represents (Dixon et al., 2018), and inadequate semantic linkages (Sequeda and Allemang, n.d.). These harms are further compounded by under-considered ad hoc data acquisition, resulting in distributional shifts, class imbalances, and hard-to-replicate phenomena (Jarrahi et al., 2023).

Quality data processing is undersupplied at present. Privatised ML value chains are ill-equipped to consider its full social value, and unincentivised to internalise the full spectrum of harms generated. Commons-based governance for infrastructural ML data, where communities could own and profit (financially or otherwise) from the data they generate, would incentivise data quality, and encode diverse values into the necessarily subjective process. DIs offer an institutional template for structuring these elements of the data commons.

Taxonomies of dataset development

By disaggregating the overlapping, multi-directional processes that constitute dataset development, I hope to map the identified technical and policy requirements of ML datasets to DI functionality. While implementation and granular design aspects of DIs are out of scope for this research, a working taxonomy of dataset development is necessary to later synthesis nine high-level principles according to the stage under considerations in the policy recommendations section.

Across the stages of dataset development, decisions are made as to what kinds of data to collect, engineer and label, as well as how to steward and distribute that data (see Jernite et al. (2022)). At present, these decisions remain mostly inscrutable and presented as purely technical considerations. However, given the harms detailed above, and the potential for 'data cascades' there is growing consensus that these decisions are neither neutral nor trivial, and that each decision point represents an opportunity for inserting greater democratic control (Sambasivan et al., 2021). DIs ought to be considerate of these decision points and designed accordingly.

There have been various overlapping taxonomies of sequencing dataset curation and their integration into the ML development process. As early as 1996, Fayyad et al. proposed nine stages. More recently, in their practitioner survey, Roh et al. (2019) articulate three phases: acquisition (discovery, augmentation, generation), labelling, and improvement (of the data itself, or the model built upon it). This informs Priestley et al's (2023) framework for *functionally grouping* tasks relating to use case and design, collection, and validation and maintenance. De Silva and Alahakoon (2022) propose a chain of dynamic data *requirements* to ML pipelines, from problem formulation, collection, cleaning, and annotation, to model training, evaluation and deployment, and inference. Other frameworks exist. All are idealised unidirectional flows of information, whereas the reality is messy iteration (Munappy et al., 2022). I draw on Priestley et al's framework in the policy recommendations below.

Much of this field of research comes from MLOps-oriented practitioners focused on identifying and disseminating best technical practice. Against this, a subset of papers consider encoding public values into technical data management decisions (Leavy et al., 2020; Priestley et al., 2023), or identify labour abuse and exploitation along the supply chain (Paullada et al., 2021). Similarly, climate policy concepts, like the 'Twin-Transition' analyse the physicality of the data cycle vis-a-vis carbon generation at each stage, and other climate concerns (see Rolnick et al. (2019)). For DIs the relevant framework depends on their mission statement.

Chapter 3

Understanding Data Institutions

Bottom-up DIs are a specific form of commons-based data governance, often predicated on a fiduciary or delegated relationship between beneficiaries and their representatives (Delacroix et al., 2020). As a parsimonious institutional template, these DIs offer an extensible model of democratic data governance. I consider DI use cases narrowly applied to mitigating existing harms and generating public value throughout the stages of ML dataset development.

The crux of this thesis (covered next section) seeks to extend the set of proposed DI use cases beyond those already well-theorised by the institutional literature's emphasis on DIs' *upstream* institutional logic (covered this section). This upstream logic primarily revolves around redistributing value across imbalanced digital markets. Additionally, I propose that DIs offer further dimensions of ML-specific utility, which become apparent when incorporating technical and policy-based critiques of current ML dataset development stemming from other fields of study.

Current redistribution-focused DI use cases include: reappropriating value extracted from disempowered data producers (Birkinbine, 2018; Loi et al., 2023; Spiekermann et al., 2021), opening up new pockets of value across the digital economy to encourage greater data sharing (Stiglich, 2023), structuring a digital commons (Delacroix and Lawrence, 2019; P. J. Singh, 2019), or supporting data availability for impactful value-aligned projects (Baarbé et al., 2019). These remain important aspects of DIs' utility to the ML community, but I further propose that establishing dedicated DIs for the design, development and stewarding of ML datasets creates integrative opportunities for both upstream *and downstream* value. To make this case, in the next section I draw on knowledge and critiques generated by the data management and (critical) global governance literatures to additionally emphasise DIs' value proposition to the ML community, impacted stakeholders (including data subjects) and wider society. But first, a genealogy of DI theorising to illustrate that just as ML datasets represent strategic policy objects, so DIs represent a strategic institutional template for dataset stewarding.

Intellectual foundations

DI theorising starts with recognising the uniqueness of data as a resource, conceptualising it as a non-rival, relational public/club good, replete with informational externalities and economies of scale (in collection and inference). This lends data well to *commons*-style governance arrangements.[1] These unique properties of data serve as the theoretical foundations for developing and evaluating the institutional and legal arrangements of digital governance efforts (Mills, 2019).

DI theorising was for a while subsumed under the label of data trusts, now that relationship is reversed. Yet it remains instructive to consider the development and diversity of input into DI theorising by tracing the development of its flagship institutional form. As the most common private law instrument for fiduciary management of assets in the public interest, trusts present data theorists with a tenable instrument for post-hoc insertion of public value considerations into the governance of increasingly privatised digital infrastructures (McDonald, 2019). Most commonly, data trusts are now proposed as a response to the growing alarm around harms deriving from the emerging 'platform model' (see Birkinbine (2018), Coyle and Manley (2021), and Meadway (2020)), although this was not always the case.

As a legal instrument structured around the appointment of a steward/trustee to manage data on beneficiaries' behalf (or the rights to data) for a predefined purpose (O'Hara, 2020), data trusts have periodically been proposed as the solution to various imbalances associated with digital markets. But trusts were not always presented as a means for representing the public interest, as they often are today. Early DI proponents innovated trusts for brokering individual digital privacy risks (Edwards, 2004) and later as (libertarian-esque) platform fiduciaries (Balkin, 2020). Ironically, recent critiques of individualist notions of data rights (see Viljoen (2021)) have also led to calls for data trusts in open data projects, whereby trusts are touted

[1] Theorising around a "digital commons" represents a major governance topic in and of itself. I consider it here only to the extent that it is relevant and adjacent to discussion of DIs. The following section explicitly considers ML's relationship to the commons, but for more, see Baarbé et al. (2019), Dietz et al. (2003), Dulong De Rosnay and Stalder (2020), Grossman et al. (2016), Huang and Siddarth (2023), Meadway (2020), Ostrom (1992), Ottolia and Sappa (2022), Potts et al. (2023), Spiekermann et al. (2021), and Yakowitz (2011))

as a governance arrangement able to incentivise greater sharing and public interest leveraging of data's aggregate utility (Thereaux and Keller, 2021). Delacroix and Lawrence (2019) persuasively developed data-trusts-as-legal-vehicle theorising, calling for a trust ecosystem constituted of differentiated units, each catering to a specific subset of governance demands and customised to community values. At a systems-level, they propose this market for trusts could accommodate a range of heterogeneous data contexts. It is this 'open data' line of reasoning that now informs DI theorising more broadly, as DIs are put forward as a form of institutional redress to imbalanced capitalistic practice undermining the digital commons.

Proponents of open data propose DIs as the structuring unit upon which to collectively steward an informational commons accountable to data-generating communities and respecting their rights and interests. These proponents range from business-oriented Zarkadakis (2020), to privacy activist organisations ("Data trusts Initiative", n.d.), to Global South stakeholders (Borokini and Saturday (2021)), and social enterprises (see "Cases — Data Collaboratives", n.d.). As the domain has matured, a plethora of alternative names and overlapping institutional propositions beyond trusts have emerged, yet the key insight of delegated fiduciary responsibility as a means to leverage data's aggregative properties, remains core.

Contested frames

DIs occupy a contested space. An early proponent of public-interest DIs was the UK government, which identified data trusts as a suitable governance instrument for data sharing infrastructure in an AI economy (Hall and Pesenti, 2017). Responding in part to the backlash against the sharing of NHS data with DeepMind, the 2017 report presented an early emphasis on downstream value creation that has since waned. Since then, the public dialogue is increasingly shaped by well-financed interests looking to maximise large-scale privately-driven data sharing (as motivated the controversy around Alphabet's Sidewalk Labs - now closed). In this framing, data trusts' privacy-enhancing functions are variously imagined as atomising, anarcho-libertarian alternatives to state control of data resources, or as disingenuous big tech cover for the freer flow and private capture of information (see Bietti (2021)).

More recently, open data proponents have taken up the DI mantle, calling for the public sector to catch up and engage (GPAI, 2023a; Marcucci and Davies, 2022). The point of departure for my proposed use of DIs for ML datasets are the variants of this discourse that frames DIs as solving the *paradox of openness* posed by extractive platforms (Jameson, 2019; Verdegem, 2022), and providing *the* structuring institution for the digital commons as a shared intellectual inheritance (Khong and Mon, 2023). In this 'open data' framing, a sustainable AI economy requires addressing the associated harms, unequal distributions of gains, and concerns about future precarity. DIs offer one such institutional means to do so; as the foundation for AI governance, inclusive data governance will increasingly form a bedrock social contract (Verhulst et al., 2023).

Beyond Tarkowski and Warso's March 2024 contribution, there have been few existing initiatives to articulate the use of collective data institutions for ML datasets, although several works allude to the possibility (see Delacroix et al. (2020), Huang and Siddarth (2023), and Zygmuntowski et al. (2021)). The closest is Chan et al. (2023), but their focus is *national*, while Jernite et al., 2022 propose a data stewardship organisation to establish and formalise relationships between actors in data ecosystems at the *international* level. On the contrary, my focus is the *community* level.

Chapter 4

The Case for Data Institutions; closing AI's *paradox of openness*

At present, private firms and AI developers exploit *open AI* discourse and access to the digital commons to deploy publicly available knowledge, tools and data to train models which are then packaged into commercial products. By not returning value to those who generated the data/knowledge, or contributing to the preservation and enrichment of the digital commons, it is depleted (from OpenFuture (2024), in line with Ostrom (1992)). This is AI's *paradox of openness*, made possible by the asymmetrically closed nature of ML development.

I propose that dedicated DIs for ML datasets close this paradox. Considering a range of technical and policy requirements for ML datasets (from other AI-related literatures), it is possible to give a *fuller* account of the dimensions across which DIs for datasets sustain and enrich the digital commons, as they return financial value to data producers, support pro-social or governance function, and offer additional services. In addition, by introducing mechanisms for democratic accountability along the data value chain, DIs open up novel access points into the inner workings of commercial ML development, incorporate a commitment to a diversity of public values by design, and create value for ML data users in a manner that regulation of models and their outputs *a priori* cannot. This downstream value proposition offsets the increased costs to be potentially faced by data users licensing these datasets. This creates room for manoeuvre in the reassignment of value towards the true costs and benefits around ML datasets. This internalising mechanism moves to close the *paradox of openness*.

ML's relationship to the digital commons

To unpack how DIs for ML datasets close AI's *paradox of openness*, first we must establish the relationship between ML systems and the digital commons.

Commons as source of representation for ML systems

Continued AI/ML progress is *dependent* on the ongoing health of the digital commons - that is the cumulative digital resources (text, media, code, legal documents, tools) collectively created, shared and managed, with established rules for access and use (Meadway, 2020). Most immediately, open-science norms around the distribution of model architectures, together with poor IP protections for software (often considered functional information, over creative expression) result in many models being made freely available over GitHub or Hugging Face as part of a digital commons. However, the dependency is more fundamental still.

The wealth of high-quality information in the digital commons is a principle source of training data for ML systems. "There is no AI without data" (Gröger, 2021; Snaith, 2023) and there is no AI without *open* data. Analytically prior to distributional claims as to how the benefits of AI ought to be allocated (to which a commons model has much to say), a robust digital commons is *necessary* to sustain present and future ML development. Recent advances in the field of AI research are a function of increasingly comprehensive and granular representations of our physical and social worlds in machine-readable format - *data*. As increased understanding of empirical scaling laws motivate the use of ever larger amounts of training data (see Hoffmann et al. (2022), Kaplan et al. (2020), and Villalobos et al. (2022)), it is clear that AI systems will continue to draw on the data and knowledge-schema of the commons for representation.[1]

AI threatens the commons

While AI progress is wholly dependent on a continuously healthy commons, the *paradox of openness* raised by current modes of AI development also threatens its viability. At a technical level, generative AI's ability to produce synthetic data at scale threatens to pollute the commons (see Ovadya and Whittlestone (2019)). At an institutional level, curating and maintaining the digital commons as a 'shared inheritance of mankind' (Khong and Mon, 2023) is a sisyphean collective endeav-

[1]For technical discussions of the development of an 'AI-ready' semantic web, see Berners-Lee (2006) and Sequeda and Allemang (n.d.); also from a management perspective see Criado and Gil-Garcia (2019)

our. Unilateral extraction of value and negation of licensing and IP norms by ML developers destabilises necessary incentives underpinning the commons' delicate self-propagation, degrading quality and paving the way for possible collapse (Meadway, 2020).

Recognising AI governance as existing on a continuum with data governance (Verhulst et al., 2023), I draw on existing accounts of digital capitalism's extractive logics to provide the contextual dynamics into which AI systems are being introduced (Jameson, 2019). From surveillance capitalism (Zuboff, 2019), and the 'data-ification' of social interactions (Couldry and Mejias, 2020; Ricaurte, 2019), to the sheer physicality of data infrastructures (Inverarity et al., 2023; Png, 2022), redressing the tendencies of digital markets for capturing commons resources has long-formed the backbone of data governance theorising, and appropriate institutional fixes have been roundly considered. AI governance scholarship is not emerging in a vacuum.

Ricaurte encapsulates the normative ambition of this branch of scholarship in describing the need to address "data-centred economies" which foster extractive models of resource exploitation, human rights violation, cultural exclusion and ecocide, concluding that "data-extractivism assumes that everything is a data source... life itself is nothing more than a continuous flow of data" (Ricaurte, 2019, p. 352). AI-enabled business models are set to exacerbate these dynamics. Digital theorists have already identified initial signs of decay within the digital commons as AI developers externalise the costs of generating data while capturing its economic value (Chan et al., 2023), or as traffic is redirected towards 'walled gardens' (Huang and Siddarth, 2023).

AI governance should leverage lessons learnt from data governance, starting with DIs. Initially theorised to rebalance digital capitalism's frictionless consumption of the commons, DIs for ML datasets could address these inequality and degradation dynamics built into current modes of training AI systems.

AI indexes the commons

Cutting against conceptions of AI as commons parasite, the third dimension of this relationship is a vision of AI systems as the digital apotheosis of the commons' current form. AI in this techno-utopian view approaches the summation of human knowledge indexed by algorithms into a 'blurry Jpeg' of the internet and query-able by chat boxes (Chiang, 2023).

As debatable (and out of scope) as that may be, a positive vision of AI systems as the inheritors of the digital commons needs to first address the *paradox of openness*, such that AI development does not bite the human hand that feeds it. DIs for ML datasets do this.

How data institutions address AI's *paradox of openness*

As strategic objects in ML supply chains, dataset governance gets to the heart the *paradox of openness*. Treating ML datasets as a commons provides a sustainable framework for balancing data use, the rights and interests of data producers, and optimum allocation of public goods across four (overlapping) dimensions. DIs could incorporate a dataset-as-a-service approach alongside community accountability and input to deliver higher quality *datasets*, as well as higher quality data *usage* around risks, liability, licensing and consent. At a governance level, the distributed leverage of an ecosystem of DIs would empower and incentivise creators, promoting greater *sustainability*, alongside *equity and inclusion*. In this way, DIs as commons-based data governance institutions incorporate the supply of public goods into the ML value chain, offset by creating new dimensions of value for downstream users.

Data quality

Beyond DIs as a resource-to-infrastructure (data-to-dataset) layer upon which to build the goods and services of an AI economy, I propose that their continuous and iterative provision of *quality* datasets is infrastructural to a healthy digital society and body politic. Delivering quality is demanding work - mostly unnoticed, until it isn't. Without adequate opportunities for scrutiny and accountability, ML data markets under-allocate quality.

There exists an extensive literature as to what quality in data means, for whom, towards which application, and how its dimensions can be measured and managed (Gudivada et al., 2017; Kreutzer et al., 2022; Munappy et al., 2022; Sambasivan et al., 2021), I operationalise quality datasets as a function of robust data work, being both *safety enhancing* (debiased, de-risked, well-sampled, coherently licensed, consensual, audited, etc) and *ML-ready* (methodically annotated, balanced across classes, low bias, feature-rich, well-structured, etc) - a composite definition drawn from Paullada et al. (2021) and Gudivada et al. (2017). Quality is value-agnostic (as is the DI institutional template) since the 'appropriate values' are use case dependent, and quality provides value both to upstream communities, and to downstream users (developers as well as future data subjects).

The expected quality gains derived from participation in open data projects is an area of active empirical research (Carey-Wilson and Stiglich, 2023; Criado and Gil-Garcia, 2019; Thereaux and Keller, 2021). Open data *in principle* allows for a representative range of social values to be incorporated and for community feedback and iteration, but also risks data pollution, value divergence, and community neglect (A. Singh and Hardinges, 2022). DIs navigate this dilemma by providing institutional templates to democratically establish context-appropriate practices for data stewardship. Specifically, curating quality datasets is a difficult, expensive and thankless process - even more so the next generation of rigorously labelled, metadata rich, ML-ready datasets (Oala et al., 2023). The true social cost of data work is simultaneously expensive and undervalued. ML developers are incentivised to free-ride on quality to capture market share, such that quality is under-allocated by the market. This results in incomplete, noisy datasets characterised by representational harms, biases, poor encoding, and imbalanced distributions (Gudivada et al., 2017).

Ultimately, companies are not designed to arbitrate public good in politically-contested spaces at scale. Companies are a legal fiction to avoid liability and maximise shareholder value. The qualities society requires of its digital infrastructure do not necessarily align with those prioritised by private firms, and analyses of data supply chains find compromised quality, obfuscated traceability and routine accountability-avoidance (Longpre et al., 2023). As artefacts functionally-dedicated to supplying the governance and technical-infrastructure for developing quality datasets, DIs offer the possibility to embed community-principles and selected public values into the design of the datasets they steward. ML data management practitioners emphasise that identifying and funding development of safety-enhancing and ML-ready datasets - among other human-design principles (see Bai et al. (2022)) - requires ongoing consultation with diverse communities, impacted stakeholders, and experts across disciplines (Aragon et al., 2022), as well as transparency and ongoing ethical consideration (Jo and Gebru, 2020). DIs provide ideal fora to facilitate the necessary engagement. In this sense, DIs double as quality-guarantors that a dataset is high-quality, human-generated, and suitable for its stated purposes.

Risk, liability, licensing, consent

Currently ML developers freely extract value from openly available commons data without adequately pricing-in associated harms, taking liability or offering recourse. This is particularly true of web scraping publicly available informational sources. The result is the externalisation of the risk and social costs associated with ML development alongside the private capture of commons value. DIs close this dimension of the *paradox of openness* by incentivising better data usage.

Compliance, liability & delegated licensing

Building on the emphasise on safety and internalisation of data's true costs in the definition of quality above, DIs assuming responsibility for collecting the consent of impacted communities and individuals, implementing necessary privacy protections, and able to guarantee regulatory compliance, would represent a gain for ML data users (Chan et al., 2023).

A network of representative stewarding institutions empowered to manage and license data to commercial ML developers in a manner consistent with the consent provided, would be further well-positioned to provide clarity over licensing options for divers ML use cases. The economics of licensing data for training LLMs and other sophisticated data-intensive modes remains intractable and beyond scope, but DIs provide an inclusive basis from which to begin an informed public discussion. Further, by collectively offering a gradation of tailored policy options differentiated by risk tolerance or other domain-specific sets of preferences, the 'market for DIs' allows individuals to select that which is most appropriate to their individual circumstances (Hardinges and Keller, 2021b). In aggregate, this sustainably incentivises greater data sharing and alleviates a source of friction in AI development.

A related constraint when opening datasets for ML training is the need to include harmful content for model learning (to be able to classify such content as harmful in newly sampled data). It remains problematic to make known harmful content open-access; DIs offer a partial solution. As a function of their proximity to the data-generating community, DIs could manage a separate category of data, evaluating toxicity (Gehman et al., 2020) and truthfulness (S. Lin et al., 2022) as it pertains to the community impacted. While not a complete solution, there's no free lunch in statistics or policy. Collective data governance can maximally balances accountability and control over harmful content against greater sharing, with a legitimacy that current modes of ML dataset curation lack.

Supporting opt-out mechanisms

DIs could better support opt-out mechanisms for privacy, benefitting both data subjects and model developers' business models, risk and approval ratings.

At present, data creators cannot prevent AI developers from using their data. Firms are by design minimally accountable under the law, opt-out mechanisms are lacking, and training datasets are closed, such that even if an individual were to threaten to withhold their data from a model developer, they lack knowledge (as to who is using their data) and bargaining power (Chan et al., 2023).

In contrast, direct community accountability incentivises DIs to facilitate individuals' data-forgetting rights by promoting technical traceability, provenance documentation and opt-out mechanisms in the datasets they steward (see Chan et al. (2023) for technical proposals). Further, as contracting organisations DIs can *license* the data they steward. When licensing data, DIs could require transparent processes from model developers for removing the influence of individual data points (see Nguyen et al. (2022)), or provide a coordinating hub for the erasure of an individual's data across all models where it is used. DIs thus empower producers to grant permission for their data to be used in one context but not in another, and to determine the after-life of the content to which they contribute.

Sustainability

Data-centric AI scholarship calls for data to be reconceptualised as an ever-evolving component in an AI system, to be iteratively augmented and developed across the system's lifecycle (Jarrahi et al., 2023). This view emphasises that ML developers require dependable supply chains providing up-to-date high-quality data at scale, continuously. Meeting this demand for data sustainability requires managing the delicate balance between openness and safeguarding the rights and interests of contributors. This is not-possible with blunt binary-choice *open AI* policy tooling.

DIs offer gradations of openness and return value (financial or other) to data producers. By realigning dataset development incentives closer to the needs of data-generating communities, DIs create more than just a legal justification and infrastructure for high-volume data sharing, rather they represent organisations worthy of societal trust (McDonald, 2019). It is trust that sustains the digital commons, and the AI systems it supports. DIs thus move towards closing this dimension of AI's *paradox of openness* by offering a governance model for sustainable data sharing infrastructure.

Equity & inclusion

The final harmful dimension associated with AI's *paradox of openness* that I propose DIs could be leveraged to address, is the reality that dominant global discourses around openness and inclusion achieve contrary aims. This happens through satisficing and non-representative inclusion. Here I draw on divers critical literatures (such as post/decolonial computing, digital divide research, data justice, indigenous data sovereignty, and cultural pluralism in AI - among others) to propose that a decentralised network of bottom-up DIs offer constructive solutions to the open questions and critiques raised by these research programmes.

The paradox of participation within *inclusive AI*

Narrow *open AI* proposals focus on making AI technologies and resources available to a broader audience, alongside facilitating collaboration and the free exchange of knowledge and data. This preoccupation with access can frustrate democratic outcomes, as unstructured access to AI systems prevents societies from restricting those uses they deem undesirable (Chan et al., 2023). Further, in practice developers instrument *open AI* discourses to maintain open access to the commons while expropriating and repackaging communal value (Gray-Widder et al., 2023). It is not enough to simply make AI products freely available (OpenFuture, 2024). Parallel to this *open AI* dynamic, paying lip service to *inclusive AI* discourses (of which there are many variants) can similarly result in exclusion, if not accompanied by intentional efforts to include underrepresented groups (Png, 2022).

Openness (*open AI*) and numerical inclusion (*inclusive AI*) alone will not dismantle existing power imbalances in AI governance. Instead policy makers must by attentive to the structures of power and systems of values that are imprinted onto technology's design and development, and be intentional as to whose representation gets included within that process. Tools are not value-agnostic, and the view from nowhere is nowhere to be found. Democratising AI involves dismantling and reassembling mechanisms (technical, institutional, or otherwise) within AI development practices that perpetuate inequalities, and DIs offer a template for institutionalising these considerations.

Divers critical literatures highlight the *paradox of participation* at the heart of global AI governance (House, 2021). Critics variously point to the reality of its domination by Global North institutions (Jobin et al., 2019; Png, 2022), its neglect of local knowledge, cultural pluralism and the demands of global fairness (R. Singh, 2021), its failure to develop adequate protocols to redress capture by vested interests (Ulnicane et al., 2021), and to processes of 'predatory inclusion' at the all levels of global governance - intended for procedural, numeric, and promotional purposes (Bliss and Neumann, 2008). Hierarchy-preserving multi-stakeholder governance initiatives such as GPAI, UN Tech Envoy Office, UNESCO, WEF, IEEE etc, are seen to be proffered in lieu of structural change, driving the *paradox* wherein inclusion can exist while structural harms persist (Png, 2022). This 'violence of care' is intended to satisfice disenfranchised stakeholders and minimise friction in advancing the status quo-preserving trajectory (see Cleaver (1999) for initial theorising). The result of this governance dysfunction are classes of learning algorithms trained on London's traffic flows or public health datasets, that with some contextual fine-tuning are ready to bring yet more benefit to Washington or Sydney-based stakeholders, yet which remain uninformative for the streets or hospitals of Kampala or Jakarta where material conditions diverge abruptly (Smith and Neupane, 2018). The cycle continues and *inclusive AI*-style rhetoric perpetuates the further bifurcation of an 'AI Divide' (Lindroth and Sinevaara-Niskanen, 2017).

DIs offer a possibility for disenfranchised stakeholders to resist this dynamic of AI systems compounding long-standing geopolitical inequalities (Couldry and Mejias, 2020). As a function of their grassroots representation of data-generating communities and the leverage they offer for withholding data, DIs give credence to calls for localised AI regulation and development operationalised to their immediate cultural and infrastructural contexts (Kak, 2020).

Digital Sovereignty

In contrast to *inclusive AI*-style optimism around 'leapfrogging' and 'digital transformation' (Borokini and Saturday, 2021; P. J. Singh, 2018), Global South digital development concerns frequently foreground digital sovereignty. DIs in this context serve as possible vectors of resistance to perceived data colonialism by institutionalising self-representation on communities' own terms.

Heterogeneity defines post-colonial scholarship. Nevertheless, Global South concerns around AI coalesce around concerns about data capitalism/colonialism (see Couldry and Mejias (2020), Graham (2014), and Ricaurte (2019)), about dependency dynamics (deriving from standard organisations' regulatory monopolies, unequal Public-Private Partnerships, or global IP regimes) (Png, 2022), and concerns about harms associated with the labour and material supply chains of AI technologies (Inverarity et al., 2023). These concerns have led to documented core-periphery patterns of under-representation and data availability in ML datasets (Graham, 2014), leading to distributional shifts, generalisation errors (even conceptual invalidity), and contextual incompatibilities in imported AI systems (Smith and Neupane, 2018).

As grassroots democratic institutions capable of withholding the means of AI production, DIs allow communities to dictate their own locally-appropriate patterns of engagement with globalised ML value chains. In the context of the Global South, this empowerment could encourage greater data sharing amongst long-suspicious populations (see Olorunju and Adams (2023) for a theoretical proposal, and Baarbé et al. (2019) for case study application to the field of agriculture). To the data management literature's concerns, DIs support intentionally collected datasets rooted in their original contexts, distributed in ways that respect the privacy, property and rights of data creators, and constructed in conversation with impacted stakeholders or domain experts (Paullada et al., 2021). Taken as a whole, DIs and the digital sovereignty they support chart a much-needed inclusive path forward for AI governance writ large.

"Nothing about us, nothing without us"

DIs support the participatory action principle that no policy should be developed without the direct participation of affected communities (see McDowell and Vetter (2023)). As representation institutions, DIs centre the knowledge of those communities most exposed to risks, and provide accountability fora for impacted stakeholders to scrutinise and surface hitherto unrecorded design decisions in ML datasets (Frey et al., 2020). In this way, DIs institutionally embody core 'participatory AI' principles that seek to involve a wider range of stakeholders than just technology developers and product risk assessors (see Ortega et al. (2020) for an agenda). DIs provide a template for institutionalising guidance as to what ML tasks (social or technical) are valid to a given community, and the infrastructure to enable creation of datasets to support those tasks.

Further, by supporting grassroots democratic participation in the ML development process, DIs could assemble communities structured by shared interests and provide a mechanism for their preferences to be aggregated into leverageable representation across higher-level advocacy platforms. Rather than subsuming stakeholders into existing top-down global governance structures without making the necessary adjustments to enact meaningful change (*inclusive AI*), DIs build from the bottom-up, accounting for heterogeneous requirements along the way.

In this way DIs could serve as an institutional response to *reality-centric AI* critiques put forward by van der Schaar (2023). This view advocates for datasets that not only advance ML as a field, but contribute positively to society. Advocates highlight the liberatory potential of intentional dataset design to make visible diverse patterns of injustice in the world, such that they may be addressed (see "Data4BlackLives" (n.d.) for an example of reality-centric dataset development). DIs for ML datasets institutionalise this line of reasoning.

Chapter 5

Policy Principles

This research focuses on the public value case for dedicated DIs stewarding ML datasets. Detailed discussion of the implementation and design of the DIs themselves is beyond scope. Nevertheless, certain high-level principles (rather than prescriptions) can be derived for policy-makers to consider.

I draw on Priestley et al's (2023) framework for the stages of the ML data lifecycle to structure these recommendations, but first I outline two stage-independent principles foundational to DI theorising.

Principle 1. ***No one-size fits all***

Data is not a homogeneous concept, rather it comes in different forms (copy-righted, anonymised, public-statistical, administrative, personal, industrial, research, transactional, health, digital media, etc). As such, data is governed by different, often overlapping, legal, contextual and custodial frameworks. Accordingly, DIs should inhabit a world of small models, customised to community values as appropriate.

Principle 2: ***Be democratic***

DIs should serve and represent a defined constituency - usually the data-generating community and impacted stakeholders (data subjects, domain experts, etc). The most explicit of these fiduciary templates is the data trust, but a range of permutations exist within the DI universe. The appropriate template will be contextually-dependent.

Use case & design

Principle 3: ***Assume a task-community role***

This extends the concept of ML 'tasks' and 'task-communities' from purely technical affairs, to also encompass societal and policy considerations.

Data availability constrains modelling efforts towards tasks on which inference can be run (Leavy et al., 2020). Accordingly, DIs should be structured around community-identified problems to support the development of purpose-built datasets tailored to real-world problems (as tasks), and ensure the result aligns with intended use.

This requires an interventionist approach, whereby DIs actively acquire data representative of the population or task at hand. This could be achieved through internal data generation, soliciting opt-ins and donations (see *Te Hiku Media*'s voice recordings drive (Nast, 2021)), or scraping data from the commons. In contrast to the current norm for minimally-supervised collection methods and crowdwork, DIs should house domain expertise and institutional specialisation to supervise collection and evaluation of gathered data for the scoped project (Jo and Gebru, 2020).

Principle 4: ***Ensure Fair Work***

Human data labelling presents the primary bottleneck to quality data, and is where most labour abuses occur (Miceli et al., 2020). DIs should provide sites of accountability and coordination for guaranteeing investment in quality data work. This should involve domain experts, community-members and consultation with impacted stakeholders (and where appropriate, semi-supervised learning).

While most commons-based efforts do not rely on paid labour, where appropriate DIs might be designed to incorporate sustainable financing and standards for ensuring fair working conditions (Tarkowski and Warso, 2024). Existing examples include Wikimedia's Enterprise API, Mozilla's Common Voice, and GIZ's FAIR Forward programme.

Collection

Principle 5: ***Promote transparency norms***

DIs should invest meaningful resources into documentation to improve understanding of datasets' contextual validity. This might include how the data was created, collected, processed, and annotated. This is essential to support informed AI accountability discussions, as well as comprehension of model capabilities/limitations, collaborative data refinement, and identification of biases (Bommasani et al., 2024). For ML developers, provenance documentation supports informed decision-making around the appropriate uses for a given dataset, and the ability to repeat the collection process to generate alternative datasets with similar characteristics, extend a dataset, or audit an experiment in a different context (Jo and Gebru, 2020; Paullada et al., 2021). Developing documentation 'in-house' allows data-generating communities (especially marginalised populations) to self-articulate the relevant contextual knowledge and downstream specifications for using their data, and/or delegate that process to appointed experts.

This represents a move towards assessing quality not only as a function of the data itself, but also the conditions, knowledge and interests of those who generated, collected, and processed the data. The datasheet for the Pile by Eleuther AI is an example of good practice.

A range of tools for transparency and accountability might be relevant. Gebru et al. (2021) proposed a standardised approach and the EU AI Act's transparency obligations represent another coordination effort, other options include data nutrition labels (Chmielinski et al., 2022), statements, and checklists.

Cleaning and (initial) preprocessing

Principle 6: *Smart quality*

DIs should incorporate quality assurance and validation tools, as well as semantic and other functionality into a dataset-as-a-service offering. Identifying noise, inconsistencies, redundancy, completeness, bias, poisoning, synthetic elements, personal identifiers, and illicit or explicit information is valuable to both data producers and users. This could be offered through auditing and structured-access protocols (Shevlane, 2022). Toolkits such as *AI Fairness 360* (Bellamy et al., 2018) provide guidance on considering protected characteristics, and checklists can document such information for legal and ethical compliance. Optional further preprocessing could consist of careful use of under/oversampling, specialised-imputation and synthetic enrichment (if transparently documented and reversible by ML developers).

Other quality enhancers include offering linked data and data integration functionality through enriched semantic metadata. This would support interoperability, federated learning, and smart contracts for data integrity and traceability. Some

DIs might consider offering a full data science SaaS product, by providing support services alongside cyberinfrastructure that co-locates data, storage, and computing infrastructure with commonly used analytics tools (see Grossman et al. (2016) for detailed technical suggestions).

Validation & maintenance

Principle 7: *Share as much as possible while respecting the decisions of data subjects*[1]

DIs should follow both open data CARE Principles For Indigenous Data Governance ("CARE Principles", 2018), and their complement - the FAIR principles - focused on data reusability ("FAIR Principles", 2016). Delivering on these principles means offering secure data storage, availability, retrieval and purging mechanisms between trusted parties, as well as protecting datasets from unauthorised access.

DIs should consider a spectrum of access regimes, selecting the most appropriate for their use case. A DI might offer multiple differentiated access modes to a single dataset. Differentiating access by user would help lower barriers to market entry by providing non-commercial users with (more) open access, subsidised by enterprise user fees. IP and personal data rights are the two most important factors to consider when designing access regimes.

In some cases an open-access commons will create most public value. In such instances making data available over existing platforms (Kaggle, HuggingFace or OpenML offer standardised data-loading infrastructure) would allow DIs to focus on providing value elsewhere. In others cases, more structured forms of commons governance, such as gated-access and active mechanisms for ensuring responsible and equitable use will be more appropriate. Here DIs would play an additional evaluative role - registering requests for access, monitoring use, and assessing impact. In these cases, existing platforms might offer sufficient control and customizability (Hugging Face offers gated-access options), otherwise dedicated sharing infrastructure may be necessary. Additionally, customised licensing and ToS design could help steer downstream use - consider *Te Hiku*'s bespoke licence combining open sharing with communal control (Nast, 2021).

Principle 8: *Sustainability*

DIs should consider structuring incentive mechanisms and perhaps even offering

[1] composite amended from Tarkowski and Warso (2024)

support for data producers and stewards. Providing the resources (infrastructure, compute, human) to design, develop, and manage quality datasets requires sustainable funding. Revenue streams from licensing data to commercial developers, or organisational teams dedicated to securing funding and grants are options. Licensing data especially generates recurring revenue, and is compatible with the CARE principles' concept of stewarding.

Principle 9: *Leverage agency through technical enforcement*

(Especially DIs handling sensitive or safety-enhancing data) should consider incorporating technical enforcement mechanisms, such as verification systems and proof-of-learning frameworks to ensure model developers are using DI data in an authorised and compliant manner (Jia et al., 2021). DIs should employ digital signatures, data poisoning and digital proofs to ensure ML developers who obtain DI data train the model with that data as agreed (see Cinà et al., 2023; Tian et al., 2022), accompanied by hash-matching to ensure the deployed model is the DI-verified one (see Chan et al. (2023)). To incentivise developers to use DI data instead of independent web-scraping or other, DIs should consider certification (similar to fair trade labels) to leverage end-user purchasing power (Chan et al., 2023).

Other compliance-supporting design choices involve allocating sufficient funding for maintaining or consulting legal and ethical counsel. Technical mechanisms can support enforcement, but assuming liability for mission-aligned or legal compliance involves developing and advocating for data sharing frameworks that account for state regulation and differences between jurisdictions, as well as addressing the ways in which deprecated datasets continue to circulate. A central repository that holds information about dataset retraction and other updates has proven insufficient, researchers must be incentivised to consult such repositories. Appropriately implementing share-alike clauses, copyleft licences, and other bespoke AI licences could further ensure value generated from the commons is at least partially returned.

Further Research

This research is an initial parsing of the landscape; it leaves much unexamined. Further work should consider frameworks for the implementation and design of DIs themselves. Legal accounts of the necessary prior justifications and frameworks are a priority - they exist, but insufficiently to provide law-makers with locally-appropriate legislation-oriented roadmaps. This is especially true at the global level, and in regions of the Global South. Economic analyses to support appropriate pricing structures for licensing data will involve the unenviable task of considering how to appropriately value data and DIs' governance services. Political-economy investigations into conditions under which DIs succeed/fail, conditional on a variety of adjustable institutional and contextual parameters, would support evidence-based institutional design. Similarly, securing financial independence and protection against corporate or government capture remains under-considered, especially as core proposed functions (such as verification and data collection) are resource intensive.

At a technical level, pressing open questions revolve around how to assert control and stewarding rights over data currently generated on privately-provisioned platform infrastructure but which might reasonably be considered commons or private data. Similarly, blockchain and cryptographic tools to develop trust and enforcement are promising areas - although purely technical solutions to human problems should be implemented cautiously.

Across all these areas, mapping existing data ecosystems to better understand the contexts into which DIs might be inserted - or already are - remains vital groundwork.

Conclusion

The current *open AI* debate produces false binaries, paradoxes, and policy-failure. This research instead preoccupies itself with the *democratisation* of AI, in an attempt to re-centre the intended societal-level outcomes of AI governance, rather than clashing over prior beliefs imported from related fields of study.

'Democratising AI' involves establishing legitimate mechanisms for supporting equitable, inclusive and sustainable distributions of benefit and access around AI development, use of AI, value-generated by AI, and governance over AI systems themselves. This research evaluates DIs as a well-theorised institutionalised form of commons data governance, according to the extent to which they might be extended to facilitate AI's democratisation.

To make the case for DIs, I first unpack datasets as strategic and underappreciated policy objects within AI governance, and propose that establishing dedicated DIs to design, develop and steward datasets for ML uses (both training and benchmark) would insert a decentralised, democratic layer of community-accountability early in the ML value chain. This upstream positioning is important. Downstream system-output level accountability and decision points tend towards restrictive or distributive policy choices, whereas DIs' upstream location embeds grassroots empowerment of data-generating communities, alongside integrative creation of public 'value' (a deliberately expansive term, encompassing a range of possible dimensions for a range of possible stakeholders).

To operationalise public 'value', I chose to focus specifically on data quality, safe and responsible data usage, data-sustainability as well as equity and inclusion (other formulations would have been valid). Doing so, I showcase how DIs might address current concerns and open questions in data-centric AI governance, and to suggest how DIs might obtain buy-in from stakeholders across the policy-making process. This latter is important for designing robust and feasible policy instruments. Finally, I propose nine high-level principles for policy-makers to consider when designing DIs for ML datasets.

This research is intended to open up a new, complementary strain of AI policy thinking around dataset development, and to add a new lever for policy-makers to consider. This is not a definitive account. There are further dimensions of value to be evaluated, feasibility and implications details to be ironed out, and constraints to be attended to. It is a first pass at synthesising the knowledge of multiple relevant domains into a coherent agenda from which to extend and expand both DI and AI governance theorising.

Appendix I - Systematic Search Python Script

Example search for googleScholar_mlLegalFrameworks_03

```
import json
import pandas as pd
from serpapi import GoogleSearch
import numpy as np
import os

# APIs
serpapi_key = "key"
openai_key = "key"

# Search name
search_id = ‘googleScholar_mlLegalFrameworks_03’

# Query String
query_string = '("ml" OR "ai" OR "intelligence" OR "machine learning") AND ("
    commons governance" OR "digital commons" OR "data commons") AND ("data
    collection" OR "data preprocessing" OR "data stewarding" OR "dataset") AND (
    "legal framework" OR "legal regime")'
review_articles_switch = "1" # review articles (0=no,1=yes)

# string for embedding comparison
topic_embedding = "Topic: legal frameworks for responsible and safe machine
    learning dataset development and data stewarding, with a focus on the legal
    regimes, legal justifications, and legal requirements that need to be in
    place, focused on the global south and global AI governance level"

df_list = []

directory = 'json'
if not os.path.exists(directory):
    os.makedirs(directory)
```

```
end = 100
num = 20
for i in range(0,end,num):
    params = {
            "engine": "google_scholar",
            "q": query_string,
            "api_key": serpapi_key,
            "start": i,
            "end": end,
            "num": num,
            "as_ylo": "1990", # from year
            "as_rr": review_articles_switch
            }

    search = GoogleSearch(params)
    results = search.get_dict()

    # save search into + topic embedding for first call to file
    if i==0:
        search_info = {"search_metadata":results["search_metadata"], "
            search_information":results["search_information"], "search_parameters
            ":results["search_parameters"], "topic_embedding":topic_embedding}
        with open('json/'+search_id+'_info'+'.json', 'w', encoding='utf-8') as f:
            json.dump(search_info, f, ensure_ascii=False, indent=4)
    else:
        pass

    # save json from every call to file
    serpapi_pagination = str(results["serpapi_pagination"]['current'])

    ## create subfolder /json/'search_id' for each search
    with open('json/'+search_id+'_page_'+serpapi_pagination+'.json', 'w',
        encoding='utf-8') as f:
        json.dump(results, f, ensure_ascii=False, indent=4)
    #metadata = results["search_metadata"]

    # add results to data frame
    organic_results = results["organic_results"]
    df_list.append(pd.DataFrame.from_dict(organic_results, orient='columns'))

# unite results from all pages and flatten json substructure where needed
df = pd.concat(df_list)
```

Appendix II - EU Common Data Spaces

Appendix II will contain a brief applied analysis of the EU's *Common Data Spaces* programme, evaluating the extent to which this Commission-led initiative achieves elements of proposed DI functionality under a primarily market-oriented approach.

Bibliography

Abebe, R., Aruleba, K., Birhane, A., Kingsley, S., Obaido, G., Remy, S. L., & Sadagopan, S. (2021). Narratives and Counternarratives on Data Sharing in Africa [arXiv:2103.01168 [cs]]. *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency*, 329–341. https://doi.org/10.1145/3442188.3445897

Abid, A., Farooqi, M., & Zou, J. (2021). Persistent Anti-Muslim Bias in Large Language Models. *Proceedings of the 2021 AAAI/ACM Conference on AI, Ethics, and Society*, 298–306. https://doi.org/10.1145/3461702.3462624

Adams, R. (2022). *AI in Africa: Key Concerns and Policy Considerations for the Future of the Continent* (tech. rep.). Africa Policy Research Institute.

Afina, Y., & Lewis, P. (2023, June). The nuclear governance model won't work for AI. Retrieved January 19, 2024, from https://www.chathamhouse.org/2023/06/nuclear-governance-model-wont-work-ai

Aragon, C., Guha, S., Kogan, M., Muller, M., & Neff, G. (2022). *Human-Centered Data Science: An Introduction*. MIT Press.

Baarbé, J., Blom, M., & de Beer, J. (2019). A Proposed "Agricultural Data Commons" in Support of Food Security [Publisher: Authors]. *The African Journal of Information and Communication*, *23*, 1–33. https://doi.org/10.23962/10539/27534

Bai, Y., Jones, A., Ndousse, K., Askell, A., Chen, A., DasSarma, N., Drain, D., Fort, S., Ganguli, D., Henighan, T., Joseph, N., Kadavath, S., Kernion, J., Conerly, T., El-Showk, S., Elhage, N., Hatfield-Dodds, Z., Hernandez, D., Hume, T., ... Kaplan, J. (2022, April). Training a Helpful and Harmless Assistant with Reinforcement Learning from Human Feedback [arXiv:2204.05862 [cs]]. https://doi.org/10.48550/arXiv.2204.05862

Balkin, J. M. (2020, September). The Fiduciary Model of Privacy. Retrieved February 27, 2024, from https://papers.ssrn.com/abstract=3700087

Bellamy, R. K. E., Dey, K., Hind, M., Hoffman, S. C., Houde, S., Kannan, K., Lohia, P., Martino, J., Mehta, S., Mojsilovic, A., Nagar, S., Ramamurthy, K. N., Richards, J., Saha, D., Sattigeri, P., Singh, M., Varshney, K. R., & Zhang, Y. (2018, October). AI Fairness 360: An Extensible Toolkit for Detecting, Understanding, and Mitigating Unwanted Algorithmic Bias [arXiv:1810.01943 [cs]]. https://doi.org/10.48550/arXiv.1810.01943

Bender, E. M., & Friedman, B. (2018). Data Statements for Natural Language Processing: Toward Mitigating System Bias and Enabling Better Science. *Transactions of the Association for Computational Linguistics*, *6*, 587–604. https://doi.org/10.1162/tacl_a_00041

Bender, E. M., Gebru, T., McMillan-Major, A., & Shmitchell, S. (2021). On the Dangers of Stochastic Parrots: Can Language Models Be Too Big? *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency*, 610–623. https://doi.org/10.1145/3442188.3445922

Benfeldt, O., Persson, J. S., & Madsen, S. (2020). Data Governance as a Collective Action Problem. *Information Systems Frontiers*, *22*(2), 299–313. https://doi.org/10.1007/s10796-019-09923-z

Berners-Lee, T. (2006, July). Linked Data - Design Issues. Retrieved January 4, 2024, from https://www.w3.org/DesignIssues/LinkedData.html

Bertuzzi, L. (2024, January). LEAK: EU Commission prepares 'strategic framework' to boost AI start-ups, generative AI uptake [Section: Artificial Intelligence]. Retrieved January 29, 2024, from https://www.euractiv.com/section/artificial-intelligence/news/leak-eu-commission-prepares-strategic-framework-to-boost-ai-start-ups-generative-ai-uptake/

Bietti, E. (2021, June). A Genealogy of Digital Platform Regulation. https://doi.org/10.2139/ssrn.3859487

Birhane, A. (2022, June). Automating Ambiguity: Challenges and Pitfalls of Artificial Intelligence [arXiv:2206.04179 [cs]]. Retrieved March 4, 2024, from http://arxiv.org/abs/2206.04179

Birhane, A., Prabhu, V. U., & Kahembwe, E. (2021, October). Multimodal datasets: Misogyny, pornography, and malignant stereotypes. Retrieved January 19, 2024, from https://arxiv.org/abs/2110.01963v1

Birkinbine, B. J. (2018). Commons Praxis: Toward a Critical Political Economy of the Digital Commons. *tripleC: Communication, Capitalism & Critique. Open Access Journal for a Global Sustainable Information Society*, *16*(1), 290–305. https://doi.org/10.31269/triplec.v16i1.929

Bliss, F., & Neumann, S. (2008). Participation in International Development Discourse and Practice: "State of the Art" and Challenges. *INEF-Report*, (94). https://doi.org/10.17185/duepublico/26915

Bommasani, R., Hudson, D. A., Adeli, E., Altman, R., Arora, S., von Arx, S., Bernstein, M. S., Bohg, J., Bosselut, A., Brunskill, E., Brynjolfsson, E., Buch, S., Card, D., Castellon, R., Chatterji, N., Chen, A., Creel, K., Davis, J. Q., Demszky, D., … Liang, P. (2022, July). On the Opportunities and Risks of Foundation Models [arXiv:2108.07258 [cs]]. https://doi.org/10.48550/arXiv.2108.07258

Bommasani, R., Klyman, K., Longpre, S., Xiong, B., Kapoor, S., Maslej, N., Narayanan, A., & Liang, P. (2024, February). Foundation Model Transparency Reports [arXiv:2402.16268 [cs]]. Retrieved May 17, 2024, from http://arxiv.org/abs/2402.16268

Borokini, F., & Saturday, B. (2021). *Exploring the future of data governace in Africa - Data Stewardship, Collaboratives, Trusts and More* (tech. rep.). Pollicy. Retrieved February 26, 2024, from https://pollicy.org/wp-content/uploads/2021/11/Exploring-the-future-of-data-governace-in-Africa.pdf

Bostrom, N. (2017). Strategic Implications of Openness in AI Development [_eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1111/1758-5899.12403]. *Global Policy*, *8*(2), 135–148. https://doi.org/10.1111/1758-5899.12403

Bucknall, B. S., & Trager, R. F. (2023). *Structured Access for Third-Party Research on Frontier AI Models: Investigating Researchers' Model Access Requirements* (tech. rep.). Centre for the Governance of AI.

Buolamwini, J., & Gebru, T. (2018). Gender Shades: Intersectional Accuracy Disparities in Commercial Gender Classification [ISSN: 2640-3498]. *Proceedings of the 1st Conference on Fairness, Accountability and Transparency*, 77–91. Retrieved April 14, 2024, from https://proceedings.mlr.press/v81/buolamwini18a.html

CARE Principles. (2018). Retrieved January 24, 2024, from https://www.gida-global.org/care

Carey-Wilson, T., & Stiglich, L. (2023, June). *Exploring the value of decentralised personal data management solutions* (tech. rep.). The Open Data Institute (ODI). Retrieved January 14, 2024, from https://theodi.org/insights/reports/exploring-the-value-of-decentralised-personal-data-management-solutions-report/

Cases — Data Collaboratives. (n.d.). Retrieved May 18, 2024, from https://datacollaboratives.org/explorer.html

Centre for Data Ethics and Innovation. (2019, June). Retrieved May 18, 2024, from https://www.gov.uk/government/organisations/centre-for-data-ethics-and-innovation

Chan, A., Bradley, H., & Rajkumar, N. (2023, May). Reclaiming the Digital Commons: A Public Data Trust for Training Data [arXiv:2303.09001 [cs]]. https://doi.org/10.48550/arXiv.2303.09001

Chen, R. J., Lu, M. Y., Chen, T. Y., Williamson, D. F. K., & Mahmood, F. (2021). Synthetic data in machine learning for medicine and healthcare [Number: 6 Publisher: Nature Publishing Group]. *Nature Biomedical Engineering*, *5*(6), 493–497. https://doi.org/10.1038/s41551-021-00751-8

Chiang, T. (2023). ChatGPT Is a Blurry JPEG of the Web [Section: annals of artificial intelligence]. *The New Yorker*. Retrieved May 21, 2024, from https://www.newyorker.com/tech/annals-of-technology/chatgpt-is-a-blurry-jpeg-of-the-web

Chmielinski, K. S., Newman, S., Taylor, M., Joseph, J., Thomas, K., Yurkofsky, J., & Qiu, Y. C. (2022, March). The Dataset Nutrition Label (2nd Gen): Leveraging Context to Mitigate Harms in Artificial Intelligence [arXiv:2201.03954 [cs]]. https://doi.org/10.48550/arXiv.2201.03954

Cilauro, F. (2021, March). *Increasing Access to Data Across the Economy* (tech. rep.). Frontier Economics.

Cinà, A. E., Grosse, K., Demontis, A., Vascon, S., Zellinger, W., Moser, B. A., Oprea, A., Biggio, B., Pelillo, M., & Roli, F. (2023). Wild Patterns Reloaded: A Survey of Machine Learning Security against Training Data Poisoning. *ACM Computing Surveys*, *55*(13s), 294:1–294:39. https://doi.org/10.1145/3585385

Cleaver, F. (1999). Paradoxes of participation: Questioning participatory approaches to development [_eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1002/%28SICI%291099-1328%28199906%2911%3A4%3C597%3A%3AAID-JID610%3E3.0.CO%3B2-Q]. *Journal of International Development*, *11*(4), 597–612. https://doi.org/10.1002/(SICI)1099-1328(199906)11:4⟨597::AID-JID610⟩3.0.CO;2-Q

Core Considerations for Exploring AI Systems as DPGs (tech. rep.). (2023, July).

Couldry, N., & Mejias, U. A. (2020). The Costs of Connection: How Data Are Colonizing Human Life and Appropriating It for Capitalism. *Social Forces*, *99*(1), e6. https://doi.org/10.1093/sf/soz172

Cox, M., Villamayor-Tomas, S., & Arnold, G. (2016). Design principles in commons science: A response to "Ostrom, Hardin and the commons" (Araral). *Environmental Science & Policy*, *61*, 238–242. https://doi.org/10.1016/j.envsci.2016.03.020

Coyle, D., Diepeveen, S., Kay, L., & Tennison, J. (2020). *The Value of Data* (tech. rep.). Bennett Institute for Public Policy. Retrieved May 17, 2024, from https://www.bennettinstitute.cam.ac.uk/wp-content/uploads/2020/12/Value_of_data_summary_report_26_Feb.pdf

Coyle, D., & Manley, A. (2021). Potential social value from data: An application of discrete choice analysis. *SSRN Electronic Journal*. https://doi.org/10.2139/ssrn.3973036

Crawford, K., & Paglen, T. (2021). Excavating AI: The politics of images in machine learning training sets. *AI & SOCIETY*, *36*(4), 1105–1116. https://doi.org/10.1007/s00146-021-01162-8

Criado, J. I., & Gil-Garcia, J. R. (2019). Creating public value through smart technologies and strategies: From digital services to artificial intelligence and beyond [Publisher: Emerald Publishing Limited]. *International Journal of Public Sector Management*, *32*(5), 438–450. https://doi.org/10.1108/IJPSM-07-2019-0178

Curry, E., Scerri, S., & Tuikka, T. (Eds.). (2022). *Data Spaces: Design, Deployment and Future Directions*. Springer International Publishing. https://doi.org/10.1007/978-3-030-98636-0

D'Addario, J. (2022, April). *Mapping data ecosystems: Methodology* (tech. rep.). Retrieved January 14, 2024, from https://theodi.org/insights/reports/mapping-data-ecosystems/

Data trusts Initiative. (n.d.). Retrieved May 18, 2024, from https://datatrusts.uk/

Data4BlackLives. (n.d.). Retrieved May 21, 2024, from https://d4bl.org/

Davies, M. (2023, February). *Data assurance: What is it and why do we need it?* (Tech. rep.). The Open Data Institute (ODI). Retrieved January 14, 2024, from https://theodi.org/insights/reports/data-assurance-white-paper/

De Silva, D., & Alahakoon, D. (2022). An artificial intelligence life cycle: From conception to production. *Patterns*, *3*(6), 100489. https://doi.org/10.1016/j.patter.2022.100489

Delacroix, S., & Lawrence, N. D. (2019). Bottom-up data Trusts: Disturbing the 'one size fits all' approach to data governance. *International Data Privacy Law*, ipz014. https://doi.org/10.1093/idpl/ipz014

Delacroix, S., Pineau, J., & Montgomery, J. (2020, June). Democratising the Digital Revolution: The Role of Data Governance. Retrieved April 15, 2024, from https://papers.ssrn.com/abstract=3720208

Deng, J., Dong, W., Socher, R., Li, L.-J., Li, K., & Fei-Fei, L. (2009). ImageNet: A Large-Scale Hierarchical Image Database. *IEEE conference on computer vision and pattern recognition*.

DeVries, T., Misra, I., & Wang, C. (2019). Does Object Recognition Work for Everyone? *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition workshops.*

DiBona, C., & Ockman, S. (1999, January). *Open Sources: Voices from the Open Source Revolution* [Google-Books-ID: bjMsCKvV9I4C]. O'Reilly Media, Inc.

Dietz, T., Ostrom, E., & Stern, P. C. (2003). The Struggle to Govern the Commons. *Science*, *302*.

Dixon, L., Li, J., Sorensen, J., Thain, N., & Vasserman, L. (2018, December). Measuring and Mitigating Unintended Bias in Text Classification. Proceedings of the 2018 AAAI/ACM Conference on AI, Ethics, and Society. Retrieved April 15, 2024, from https://dl.acm.org/doi/10.1145/3278721.3278729

DNP Label Maker. (n.d.). Retrieved February 24, 2024, from https://labelmaker.datanutrition.org/

DPGA. (2023, October). Exploring a Gradient Approach to the Openness of AI System Components - Digital Public Goods Alliance. Retrieved January 9, 2024, from https://dpgalliance.github.io/blog/exploring-a-gradient-approach-to-the-openness-of-ai-system-components/

Dulong De Rosnay, M., & Stalder, F. (2020). Digital commons. *Internet Policy Review*, *9*(4). https://doi.org/10.14763/2020.4.1530

Dumas, J. (2023, July). The AI2 ImpACT License - A new way to think about AI licensing. Retrieved May 17, 2024, from https://blog.allenai.org/the-ai2-impact-license-a-new-way-to-think-about-ai-licensing-bc90ff26a9ee

Dwork, C., Hardt, M., Pitassi, T., Reingold, O., & Zemel, R. (2012). Fairness through awareness. *Proceedings of the 3rd Innovations in Theoretical Computer Science Conference*, 214–226. https://doi.org/10.1145/2090236.2090255

Eaves, D., Bolte, L., Chuquihuara, O., & Hodigere, S. (n.d.). Best Practices for the Governance of Digital Public Goods.

Edwards, L. (2004). The Problem With Privacy – A Modest Proposal. Retrieved April 16, 2024, from https://tinyurl.com/bdff4tya

Elcock, G. (2021, July). *Unlocking the value of data: Exploring the role of data intermediaries* (tech. rep.). Centre for Data Ethics & Innovation. https://assets.publishing.service.gov.uk/media/60f83e4ce90e0764c7950d09/Data_intermediaries_-_accessible_version.pdf

Eom, B., Lim, S., Suh, Y.-H., Woo, S., Park, D., & Park, C. (2021). Artificial Intelligence-Enabled Data Value Curation on AI-Data Commons [ISSN: 2162-1233]. *2021 International Conference on Information and Communication*

Technology Convergence (ICTC), 1316–1318. https://doi.org/10.1109/ICTC52510.2021.9620838

Ethical Web and Data Architectures. (n.d.). Retrieved May 18, 2024, from https://www.oxfordmartin.ox.ac.uk/ethical-web-and-data-architectures

EU AI Act: First regulation on artificial intelligence. (2023, June). Retrieved May 19, 2024, from https://www.europarl.europa.eu/topics/en/article/20230601STO93804/eu-ai-act-first-regulation-on-artificial-intelligence

FAIR Principles. (2016). Retrieved January 19, 2024, from https://www.go-fair.org/fair-principles/

Forum, W. E. (2023, October). *Data Equity Concepts - Generative AI* (tech. rep.). World Economic Forum. Retrieved February 5, 2024, from https://www3.weforum.org/docs/WEF_Data_Equity_Concepts_Generative_AI_2023.pdf

Frey, W. R., Patton, D. U., Gaskell, M. B., & McGregor, K. A. (2020). Artificial Intelligence and Inclusion: Formerly Gang-Involved Youth as Domain Experts for Analyzing Unstructured Twitter Data [Publisher: SAGE Publications Inc]. *Social Science Computer Review*, *38*(1), 42–56. https://doi.org/10.1177/0894439318788314

Gebru, T., Morgenstern, J., Vecchione, B., Vaughan, J. W., Wallach, H., Daumé III, H., & Crawford, K. (2021, December). Datasheets for Datasets [arXiv:1803.09010 [cs]]. https://doi.org/10.48550/arXiv.1803.09010

Gehman, S., Gururangan, S., Sap, M., Choi, Y., & Smith, N. A. (2020, November). RealToxicityPrompts: Evaluating Neural Toxic Degeneration in Language Models. In T. Cohn, Y. He, & Y. Liu (Eds.), *Findings of the Association for Computational Linguistics: EMNLP 2020* (pp. 3356–3369). Association for Computational Linguistics. https://doi.org/10.18653/v1/2020.findings-emnlp.301

Gelman, A., Mattson, G., & Simpson, D. (2018). Gaydar and the Fallacy of Decontextualized Measurement. *Sociological Science*, *5*, 270–280. https://doi.org/10.15195/v5.a12

Ghai, B., Liao, Q. V., Zhang, Y., & Mueller, K. (2020, April). Measuring Social Biases of Crowd Workers using Counterfactual Queries [arXiv:2004.02028 [cs]]. Retrieved April 15, 2024, from http://arxiv.org/abs/2004.02028

Google, T. P. t. (n.d.). Know Your Data. Retrieved January 19, 2024, from https://knowyourdata.withgoogle.com

Gorwa, R., & Veale, M. (2024). Moderating Model Marketplaces: Platform Governance Puzzles for AI Intermediaries [Publisher: OSF]. https://doi.org/10.31235/osf.io/6dfk3

Gosh, R. (2007). Economic impact of open source software on innovation and the competitiveness of the Information and Communication Technologies (ICT) sector in the EU. Retrieved January 13, 2024, from https://www.%20semanticscholar.%20org/paper/Economic-impact-of-open-source-software-on-and-the-Ghosh/4f0469c3702f5a22176265c72d0d764bc0447774

GPAI. (2023a). *The Role of Government as a Provider of Data for Artificial Intelligence - Interim Report* (tech. rep.). Global Public Partnership on AI (GPAI).

GPAI. (2023b, October). *Designing Trustworthy Data Institutions - Scanning the Local Data Ecosystem in Climate-Induced Migration in Lake Chad Basin - Pilot Study in Cameroon* (tech. rep.). Global Public Partnership on AI (GPAI).

Graham, M. (2014, June). A Critical Perspective on the Potential of the Internet at the Margins of the Global Economy. https://doi.org/10.2139/ssrn.2448223

Gray-Widder, D., West, S., & Whittaker, M. (2023). Open (For Business): Big Tech, Concentrated Power, and the Political Economy of Open AI. *SSRN Electronic Journal*. https://doi.org/10.2139/ssrn.4543807

Gröger, C. (2021). There is no AI without data. *Communications of the ACM*, *64*(11), 98–108. https://doi.org/10.1145/3448247

Grossman, R. L., Heath, A., Murphy, M., Patterson, M., & Wells, W. (2016). A Case for Data Commons. *Computing in science & engineering*, *18*(5), 10–20. https://doi.org/10.1109/MCSE.2016.92

Gudivada, V. N., Apon, A., & Ding, J. (2017). Data Quality Considerations for Big Data and Machine Learning: Going Beyond Data Cleaning and Transformations. *International Journal on Advances in Software*, *10*.

Hall, W., & Pesenti, J. (2017, October). *Growing the artificial intelligence industry in the UK* (tech. rep.). UK Government Department for Science, Innovation and Technology. Retrieved April 16, 2024, from https://www.gov.uk/government/publications/growing-the-artificial-intelligence-industry-in-the-uk

Han, D., Choe, J., Chun, S., Chung, J. J. Y., Chang, M., Yun, S., Song, J. Y., & Oh, S. J. (2023, July). Neglected Free Lunch – Learning Image Classifiers Using Annotation Byproducts [arXiv:2303.17595 [cs]]. https://doi.org/10.48550/arXiv.2303.17595

Hardinges, J., & Keller, J. R. (2021a, January). What are data institutions and why are they important? Retrieved May 17, 2024, from https://theodi.org/insights/explainers/what-are-data-institutions-and-why-are-they-important/

Hardinges, J., & Keller, J. R. (2021b, June). What are 'bottom-up' data institutions and how do they empower people? Retrieved January 14, 2024, from https://theodi.org/news-and-events/blog/what-are-bottom-up-data-institutions-and-how-do-they-empower-people/

Hardinges, J., & Scott, A. (2018, May). Most data is about multiple people – what does that mean for data portability? Retrieved January 14, 2024, from https://theodi.org/news-and-events/blog/most-data-is-about-multiple-people-what-does-that-mean-for-data-portability/

Harvey, A., & LaPlace, J. (2021, December). Researchers Gone Wild: Origins and Endpoints of Image Training Datasets Created "In the Wild". In *Practicing Sovereignty* (pp. 289–310). https://doi.org/10.1515/9783839457603-016

Hoffmann, J., Borgeaud, S., Mensch, A., Buchatskaya, E., Cai, T., Rutherford, E., Casas, D. d. l., Hendricks, L. A., Welbl, J., Clark, A., Hennigan, T., Noland, E., Millican, K., Driessche, G. v. d., Damoc, B., Guy, A., Osindero, S., Simonyan, K., Elsen, E., ... Sifre, L. (2022). An empirical analysis of compute-optimal large language model training. *Advances in Neural Information Processing Systems*. Retrieved April 15, 2024, from https://openreview.net/forum?id=iBBcRUlOAPR

Homepage. (n.d.). Retrieved November 23, 2024, from https://fair.work/en/fw/homepage/

House, C. (2021). *Reflections on building more inclusive global governance* (tech. rep.). Chatham House. Retrieved April 14, 2024, from https://www.chathamhouse.org/sites/default/files/2021-04/2021-04-15-reflections-building-inclusive-global-governance.pdf

Huang, S., & Siddarth, D. (2023, March). Generative AI and the Digital Commons [arXiv:2303.11074 [cs]]. Retrieved January 19, 2024, from http://arxiv.org/abs/2303.11074

Hube, C., Fetahu, B., & Gadiraju, U. (2019). Understanding and Mitigating Worker Biases in the Crowdsourced Collection of Subjective Judgments. *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*, 1–12. https://doi.org/10.1145/3290605.3300637

Inverarity, C., Moriniere, S., Snaith, B., Keller, J. R., Redler-Hawes, H., & Freeman, J. (2023, April). Power, ecology and diplomacy in critical data infrastructures. Retrieved January 19, 2024, from https://theodi.org/insights/reports/power-ecology-and-diplomacy-in-critical-data-infrastructures/

Jakubik, J., Vössing, M., Kühl, N., Walk, J., & Satzger, G. (2024, January). Data-Centric Artificial Intelligence [arXiv:2212.11854 [cs]]. https://doi.org/10.48550/arXiv.2212.11854

Jameson, S. (2019, April). Extractive Logics as a Systemic Pattern. Retrieved April 15, 2024, from https://globaldatajustice.org/gdj/1883/

Jarrahi, M. H., Memariani, A., & Guha, S. (2023). The Principles of Data-Centric AI. *Communications of the ACM*, *66*(8), 84–92. https://doi.org/10.1145/3571724

Jernite, Y., Nguyen, H., Biderman, S., Rogers, A., Masoud, M., Danchev, V., Tan, S., Luccioni, A. S., Subramani, N., Johnson, I., Dupont, G., Dodge, J., Lo, K., Talat, Z., Radev, D., Gokaslan, A., Nikpoor, S., Henderson, P., Bommasani, R., & Mitchell, M. (2022). Data Governance in the Age of Large-Scale Data-Driven Language Technology. *Proceedings of the 2022 ACM Conference on Fairness, Accountability, and Transparency*, 2206–2222. https://doi.org/10.1145/3531146.3534637

Jia, H., Yaghini, M., Choquette-Choo, C. A., Dullerud, N., Thudi, A., Chandrasekaran, V., & Papernot, N. (2021). Proof-of-Learning: Definitions and Practice. *2021 IEEE Symposium on Security and Privacy (SP)*, 1039–1056. https://doi.org/10.1109/SP40001.2021.00106

Jo, E. S., & Gebru, T. (2020). Lessons from archives: Strategies for collecting sociocultural data in machine learning. *Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency*, 306–316. https://doi.org/10.1145/3351095.3372829

Jobin, A., Ienca, M., & Vayena, E. (2019). The global landscape of AI ethics guidelines [Publisher: Nature Publishing Group]. *Nature Machine Intelligence*, *1*(9), 389–399. https://doi.org/10.1038/s42256-019-0088-2

Kak, A. (2020). "The Global South is everywhere, but also always somewhere": National Policy Narratives and AI Justice. *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society*, 307–312. https://doi.org/10.1145/3375627.3375859

Kaplan, J., McCandlish, S., Henighan, T., Brown, T. B., Chess, B., Child, R., Gray, S., Radford, A., Wu, J., & Amodei, D. (2020, January). Scaling Laws for Neural Language Models [arXiv:2001.08361 [cs, stat]]. https://doi.org/10.48550/arXiv.2001.08361

Kapoor, S. (2023, March). Is the future of AI open or closed? Retrieved January 2, 2024, from https://www.aisnakeoil.com/p/is-the-future-of-ai-open-or-closed

Kathikar, A., Nair, A., Lazarine, B., Sachdeva, A., & Samtani, S. (2023, July). *Assessing the Vulnerabilities of the Open-Source Artificial Intelligence (AI) Landscape: A Large-Scale Analysis of the Hugging Face Platform*. https://doi.org/10.1109/ISI58743.2023.10297271

Keller, J. R. (2021, September). *How do data institutions facilitate safe access to sensitive data?* (Tech. rep.). The Open Data Institute (ODI). Retrieved January 19, 2024, from https://theodi.org/insights/reports/how-do-data-institutions-facilitate-safe-access-to-sensitive-data/

Khan, L. M. (2017). Amazon's Antitrust Paradox. *THE YALE LAW JOURNAL*.

Khan, M., & Hanna, A. (2022, September). The Subjects and Stages of AI Dataset Development: A Framework for Dataset Accountability. https://doi.org/10.2139/ssrn.4217148

Khong, D. W. K., & Mon, S. W. (2023). Artificial Intelligence as a Common Heritage of Mankind [Number: 1]. *UUM Journal of Legal Studies*, *14*(1), 113–139. https://doi.org/10.32890/uumjls2023.14.1.5

Kirstain, Y., Polyak, A., Singer, U., Matiana, S., Penna, J., & Levy, O. (2023, November). Pick-a-Pic: An Open Dataset of User Preferences for Text-to-Image Generation [arXiv:2305.01569 [cs]]. https://doi.org/10.48550/arXiv.2305.01569

Koch, B., Denton, E., Hanna, A., & Foster, J. G. (2021, December). Reduced, Reused and Recycled: The Life of a Dataset in Machine Learning Research [arXiv:2112.01716 [cs, stat]]. https://doi.org/10.48550/arXiv.2112.01716

Kreutzer, J., Caswell, I., Wang, L., Wahab, A., van Esch, D., Ulzii-Orshikh, N., Tapo, A., Subramani, N., Sokolov, A., Sikasote, C., Setyawan, M., Sarin, S., Samb, S., Sagot, B., Rivera, C., Rios, A., Papadimitriou, I., Osei, S., Suarez, P. O., ... Adeyemi, M. (2022). Quality at a Glance: An Audit of Web-Crawled Multilingual Datasets [arXiv:2103.12028 [cs]]. *Transactions of the Association for Computational Linguistics*, *10*, 50–72. https://doi.org/10.1162/tacl_a_00447

Leavy, S., O'Sullivan, B., & Siapera, E. (2020, July). Data, Power and Bias in Artificial Intelligence [arXiv:2008.07341 [cs]]. https://doi.org/10.48550/arXiv.2008.07341

Liesenfeld, A., Lopez, A., & Dingemanse, M. (2023). Opening up ChatGPT: Tracking openness, transparency, and accountability in instruction-tuned text generators. *Proceedings of the 5th International Conference on Conversational User Interfaces*, 1–6. https://doi.org/10.1145/3571884.3604316

Lim, S., Suh, Y.-H., Park, D., Woo, S., & Park, C. (2020). Design of SW Framework for Trustworthy AI-Data Commons [ISSN: 2162-1233]. *2020 International Conference on Information and Communication Technology Convergence (ICTC)*, 1883–1885. https://doi.org/10.1109/ICTC49870.2020.9289370

Lin, S., Hilton, J., & Evans, O. (2022, May). TruthfulQA: Measuring How Models Mimic Human Falsehoods [arXiv:2109.07958 [cs]]. https://doi.org/10.48550/arXiv.2109.07958

Lin, T. (2012). The New Investor. *UCLA Law Review*, *60*. Retrieved March 26, 2024, from https://scholarship.law.ufl.edu/cgi/viewcontent.cgi?article=1245&context=facultypub

Lindroth, M., & Sinevaara-Niskanen, H. (2017, November). *Global Politics and Its Violent Care for Indigeneity: Sequels to Colonialism*. Springer.

Loi, M., Dehaye, P.-O., & Hafen, E. (2023). Towards Rawlsian ‘property-owning democracy’ through personal data platform cooperatives. *Critical Review of International Social and Political Philosophy*, *26*(6), 769–787. https://doi.org/10.1080/13698230.2020.1782046

Longpre, S., Mahari, R., Chen, A., Obeng-Marnu, N., Sileo, D., Brannon, W., Muennighoff, N., Khazam, N., Kabbara, J., Perisetla, K., Wu, X., Shippole, E., Bollacker, K., Wu, T., Villa, L., Pentland, S., & Hooker, S. (2023, November). The Data Provenance Initiative: A Large Scale Audit of Dataset Licensing & Attribution in AI [arXiv:2310.16787 [cs]]. Retrieved May 17, 2024, from http://arxiv.org/abs/2310.16787

Luccioni, A. S., Jernite, Y., & Mitchell, M. (2021, November). Introducing the Data Measurements Tool: An Interactive Tool for Looking at Datasets. Retrieved January 19, 2024, from https://huggingface.co/blog/data-measurements-tool

Lum, K., & Isaac, W. (2016). To predict and serve? *Significance*, *13*(5), 14–19. https://doi.org/10.1111/j.1740-9713.2016.00960.x

Maffulli, S. (2023, July). Towards a definition of ”Open Artificial Intelligence”: First meeting recap [Publication Title: Voices of Open Source]. Retrieved November 14, 2023, from https://blog.opensource.org/towards-a-definition-of-open-artificial-intelligence-first-meeting-recap/

Mann, G., & O’Neil, C. (2016). Hiring Algorithms Are Not Neutral [Section: Analytics and data science]. *Harvard Business Review*. Retrieved May 26, 2024, from https://hbr.org/2016/12/hiring-algorithms-are-not-neutral

Marcucci, S., & Davies, M. (2022, March). *How can the UK government and other policymakers support ‘bottom-up data institutions’?* (Tech. rep.). The Open Data Institute (ODI). Retrieved January 14, 2024, from https://theodi.org/insights/reports/how-can-the-uk-government-and-other-policymakers-support-bottom-up-data-institutions/

Mazumder, M., Banbury, C., Yao, X., Karlaš, B., Rojas, W. G., Diamos, S., Diamos, G., He, L., Parrish, A., Kirk, H. R., Quaye, J., Rastogi, C., Kiela, D., Jurado, D., Kanter, D., Mosquera, R., Ciro, J., Aroyo, L., Acun, B., ... Reddi, V. J. (2023, October). DataPerf: Benchmarks for Data-Centric AI Development [arXiv:2207.10062 [cs]]. https://doi.org/10.48550/arXiv.2207.10062

McDonald, S. M. (2019, March). Reclaiming Data Trusts. Retrieved February 26, 2024, from https://www.cigionline.org/articles/reclaiming-data-trusts/

McDowell, Z. J., & Vetter, M. A. (2023). Rethinking Artificial Intelligence: Algorithmic Bias and Ethical Issues— The Realienation of the Commons: Wikidata and the Ethics of "Free" Data [Number: 0]. *International Journal of Communication*, *18*(0), 19. Retrieved January 4, 2024, from https://ijoc.org/index.php/ijoc/article/view/20807

Meadway, J. (2020). Creating a Digital Commons.

Metz, C., Kang, C., Frenkel, S., Thompson, S. A., & Grant, N. (2024). How Tech Giants Cut Corners to Harvest Data for A.I. *The New York Times*. Retrieved May 27, 2024, from https://www.nytimes.com/2024/04/06/technology/tech-giants-harvest-data-artificial-intelligence.html

Miceli, M., Schuessler, M., & Yang, T. (2020). Between Subjectivity and Imposition: Power Dynamics in Data Annotation for Computer Vision. *Proceedings of the ACM on Human-Computer Interaction*, *4*(CSCW2), 115:1–115:25. https://doi.org/10.1145/3415186

Mills, S. (2019). Who Owns the Future? Data Trusts, Data Commons, and the Future of Data Ownership. *SSRN Electronic Journal*. https://doi.org/10.2139/ssrn.3437936

Mökander, J. (2021). On the Limits of Design: What Are the Conceptual Constraints on Designing Artificial Intelligence for Social Good? [Series Title: Digital Ethics Lab Yearbook]. In J. Cowls & J. Morley (Eds.), *The 2020 Yearbook of the Digital Ethics Lab* (pp. 39–52). Springer International Publishing. https://doi.org/10.1007/978-3-030-80083-3_5

Mökander, J., Axente, M., Casolari, F., & Floridi, L. (2022). Conformity Assessments and Post-market Monitoring: A Guide to the Role of Auditing in the Proposed European AI Regulation. *Minds and Machines*, *32*(2), 241–268. https://doi.org/10.1007/s11023-021-09577-4

Munappy, A. R., Bosch, J., Olsson, H. H., Arpteg, A., & Brinne, B. (2022). Data management for production quality deep learning models: Challenges and solutions. *Journal of Systems and Software*, *191*, 111359. https://doi.org/10.1016/j.jss.2022.111359

Musoke, A. A., Nishimirwe, J. P., Lawal, N., & Gueye, A. (2023). Digital Public Goods Interoperability: A Low-Code Middleware Approach. *Proceedings of* *the 6th ACM SIGCAS/SIGCHI Conference on Computing and Sustainable Societies*, 138–141. https://doi.org/10.1145/3588001.3609376

Myers, A. (2022, April). In Human-Centered AI, the Boundaries Between UX and Software Roles Are Evolving. Retrieved January 19, 2024, from https://hai.stanford.edu/news/human-centered-ai-boundaries-between-ux-and-software-roles-are-evolving

Nast, C. (2021). Māori are trying to save their language from Big Tech [Section: tags]. *Wired UK*. Retrieved January 14, 2024, from https://www.wired.co.uk/article/maori-language-tech

National Data Strategy. (n.d.). Retrieved May 18, 2024, from https://www.gov.uk/government/publications/uk-national-data-strategy/national-data-strategy

Nguyen, T. T., Huynh, T. T., Nguyen, P. L., Liew, A. W.-C., Yin, H., & Nguyen, Q. V. H. (2022, October). A Survey of Machine Unlearning [arXiv:2209.02299 [cs]]. https://doi.org/10.48550/arXiv.2209.02299

Nicholson, B., Nielsen, P., Sæbø, J. I., & Tavares, A. P. (2022). Digital Public Goods for Development: A Conspectus and Research Agenda. In Y. Zheng, P. Abbott, & J. A. Robles-Flores (Eds.), *Freedom and Social Inclusion in a Connected World* (pp. 455–470). Springer International Publishing. https://doi.org/10.1007/978-3-031-19429-0_27

Nordhaug, L. M., & Harris, L. (2021, December). *Digital public goods: Enablers of digital sovereignty* (tech. rep.). OECD. Paris. https://doi.org/10.1787/c023cb2e-en

Oala, L., Maskey, M., Bat-Leah, L., Parrish, A., Gürel, N. M., Kuo, T.-S., Liu, Y., Dror, R., Brajovic, D., Yao, X., Bartolo, M., Rojas, W. A. G., Hileman, R., Aliment, R., Mahoney, M. W., Risdal, M., Lease, M., Samek, W., Dutta, D., . . . Mattson, P. (2023, November). DMLR: Data-centric Machine Learning Research – Past, Present and Future [arXiv:2311.13028 [cs, eess]]. https://doi.org/10.48550/arXiv.2311.13028

O'Hara, K. (2020). Data Trusts [Publisher: Lexxion Publisher]. *European Data Protection Law Review*, *6*(4), 484–491. https://doi.org/10.21552/edpl/2020/4/4

Olorunju, N., & Adams, R. (2023). African data trusts: New tools towards collective data governance? *Information & Communications Technology Law*, 1–14. https://doi.org/10.1080/13600834.2023.2260678

Open Definition 2.1 - Defining Open in Open Data, Open Content and Open Knowledge. (n.d.). Retrieved January 3, 2024, from https://opendefinition.org/od/2.1/en/

The Open Source AI Definition. (n.d.). Retrieved January 9, 2024, from https://hackmd.io/@opensourceinitiative/opensource-ai-definition-0-2

OpenFuture. (2024, January). The Paradox of Open: Policies for the Digital Commons. Retrieved February 27, 2024, from https://paradox.openfuture.eu/policies/

Ortega, A., Pomares, J., & Belén Abdala, M. (2020, October). *Managing the transition to a multi-stakeholder artificial intelligence governance* (tech. rep.). Retrieved April 15, 2024, from https://www.realinstitutoelcano.org/en/analyses/managing-the-transition-to-a-multi-stakeholder-artificial-intelligence-governance/

Ostrom, E. (1992). Governing the Commons: The Evolution of Institutions for Collective Action.

Ottolia, A., & Sappa, C. (2022). A Topography of Data Commons: From Regulation to Private Dynamism. *GRUR International*, *71*(4), 335–345. https://doi.org/10.1093/grurint/ikab156

Ovadya, A., & Whittlestone, J. (2019, July). Reducing malicious use of synthetic media research: Considerations and potential release practices for machine learning [arXiv:1907.11274 [cs]]. Retrieved January 2, 2024, from http://arxiv.org/abs/1907.11274

Paullada, A., Raji, I. D., Bender, E. M., Denton, E., & Hanna, A. (2021). Data and its (dis)contents: A survey of dataset development and use in machine learning research. *Patterns*, *2*(11), 100336. https://doi.org/10.1016/j.patter.2021.100336

Png, M.-T. (2022). At the Tensions of South and North: Critical Roles of Global South Stakeholders in AI Governance. *Proceedings of the 2022 ACM Conference on Fairness, Accountability, and Transparency*, 1434–1445. https://doi.org/10.1145/3531146.3533200

Polyzotis, N., Roy, S., Whang, S. E., & Zinkevich, M. (2018). Data Lifecycle Challenges in Production Machine Learning: A Survey. *SIGMOD Record*, *47*(2). https://doi.org/10.1145/3299887.3299891

Potts, J., Torrance, A., Harhoff, D., & Von Hippel, E. (2023). Profiting from Data Commons: Theory, Evidence, and Strategy Implications. *Strategy Science*, stsc.2021.0080. https://doi.org/10.1287/stsc.2021.0080

Prabhu, V. U., & Birhane, A. (2020, July). Large image datasets: A pyrrhic win for computer vision? [arXiv:2006.16923 [cs, stat]]. Retrieved April 14, 2024, from http://arxiv.org/abs/2006.16923

Priestley, M., O'donnell, F., & Simperl, E. (2023). A Survey of Data Quality Requirements That Matter in ML Development Pipelines. *Journal of Data and Information Quality*, *15*(2), 1–39. https://doi.org/10.1145/3592616

A Public Option for AI Development. (n.d.). Retrieved February 27, 2024, from https://openfuture.eu/policies-for-the-digital-commons/a-public-option-for-ai-development

Quinn, J., Frias-Martinez, V., & Subramanian, L. (2014). Computational Sustainability and Artificial Intelligence in the Developing World. *AI Magazine*, *35*(3), 36–47. https://doi.org/10.1609/aimag.v35i3.2529

RAD Tip Sheet. (n.d.). Retrieved January 19, 2024, from https://www.theengineroom.org/wp-content/uploads/2022/05/RAD-Tip-Sheets.pdf

Radin, J. (2017). "Digital Natives": How Medical and Indigenous Histories Matter for Big Data [Publisher: The University of Chicago Press]. *Osiris*, *32*(1), 43–64. https://doi.org/10.1086/693853

Raji, I. D., & Fried, G. (2021, February). About Face: A Survey of Facial Recognition Evaluation [arXiv:2102.00813 [cs]]. https://doi.org/10.48550/arXiv.2102.00813

Raji, I. D., Smart, A., White, R. N., Mitchell, M., Gebru, T., Hutchinson, B., Smith-Loud, J., Theron, D., & Barnes, P. (2020, January). Closing the AI Accountability Gap: Defining an End-to-End Framework for Internal Algorithmic Auditing [arXiv:2001.00973 [cs]]. https://doi.org/10.48550/arXiv.2001.00973

Responsible AI Licenses (RAIL) [Publication Title: Responsible AI Licenses (RAIL)]. (n.d.). Retrieved November 14, 2023, from https://www.licenses.ai

Ricaurte, P. (2019). Data Epistemologies, The Coloniality of Power, and Resistance [Publisher: SAGE Publications]. *Television & New Media*, *20*(4), 350–365. https://doi.org/10.1177/1527476419831640

Robertson, A. (2024). Google apologizes for "missing the mark" after Gemini generated racially diverse Nazis. *The Verge*. Retrieved February 22, 2024, from https://www.theverge.com/2024/2/21/24079371/google-ai-gemini-generative-inaccurate-historical

Roh, Y., Heo, G., & Whang, S. E. (2019, August). A Survey on Data Collection for Machine Learning: A Big Data – AI Integration Perspective [arXiv:1811.03402 [cs, stat]]. https://doi.org/10.48550/arXiv.1811.03402

Rolnick, D., Donti, P. L., Kaack, L. H., Kochanski, K., Lacoste, A., Sankaran, K., Ross, A. S., Milojevic-Dupont, N., Jaques, N., Waldman-Brown, A., Luccioni, A., Maharaj, T., Sherwin, E. D., Mukkavilli, S. K., Kording, K. P., Gomes, C., Ng, A. Y., Hassabis, D., Platt, J. C., … Bengio, Y. (2019, November). Tackling Climate Change with Machine Learning [arXiv:1906.05433 [cs, stat]]. https://doi.org/10.48550/arXiv.1906.05433

Saebo, J. I., Nicholson, B., Nielsen, P., & Sahay, S. (2021). Digital Global Public Goods. *Proceedings of the 1st Virtual Conference on Implications of Information and Digital Technologies for Development, 2021.*

Sag, M. (2019). The New Legal Landscape for Text Mining and Machine Learning. *SSRN Electronic Journal*. https://doi.org/10.2139/ssrn.3331606

Sambasivan, N., Kapania, S., Highfill, H., Akrong, D., Paritosh, P., & Aroyo, L. M. (2021). "Everyone wants to do the model work, not the data work": Data Cascades in High-Stakes AI. *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*, 1–15. https://doi.org/10.1145/3411764.3445518

Scholz, T., & Schneider, N. (Eds.). (2016). *Ours to hack and to own: The rise of platform cooperativism, a new vision for the future of work and a fairer internet*. OR Books.

Schuhmann, C., Beaumont, R., Vencu, R., Gordon, C., Wightman, R., Cherti, M., Coombes, T., Katta, A., Mullis, C., Wortsman, M., Schramowski, P., Kundurthy, S., Crowson, K., Schmidt, L., Kaczmarczyk, R., & Jitsev, J. (2022, October). LAION-5B: An open large-scale dataset for training next generation image-text models [arXiv:2210.08402 [cs]]. https://doi.org/10.48550/arXiv.2210.08402

Seger, E., Dreksler, N., Moulange, R., Dardaman, E., Schuett, J., Wei, K., Winter, C., Arnold, M., hÉigeartaigh, S. Ó., Korinek, A., Anderljung, M., Bucknall, B., Chan, A., Stafford, E., Koessler, L., Ovadya, A., Garfinkel, B., Bluemke, E., Aird, M., ... Gupta, A. (2023, September). Open-Sourcing Highly Capable Foundation Models: An evaluation of risks, benefits, and alternative methods for pursuing open-source objectives [arXiv:2311.09227 [cs]]. Retrieved January 2, 2024, from http://arxiv.org/abs/2311.09227

Sequeda, j., & Allemang, D. (n.d.). WTF is a Knowledge Graph and Why is it Key to AI? Retrieved January 19, 2024, from https://data.world/resources/webinar/wtf-is-a-knowledge-graph-and-why-is-it-key-to-ai/

Shadbolt, N. (2022). "From So Simple a Beginning": Species of Artificial Intelligence. *Daedalus*, *151*(2), 28–42. https://doi.org/10.1162/daed_a_01898

Shahbazi, N., & Asudeh, A. (2022, April). *Data-Centric Distrust Quantification for Responsible AI: When Data-driven Outcomes Are Not Reliable.*

Shevlane, T. (2022, April). Structured Access: An emerging paradigm for safe AI deployment [arXiv:2201.05159 [cs]]. https://doi.org/10.48550/arXiv.2201.05159

Shrestha, Y., Krogh, G., & Feuerriegel, S. (2023). Building open-source AI. *Nature Computational Science*, *3*. https://doi.org/10.1038/s43588-023-00540-0

Shteyn, A., Kollnig, K., & Inverarity, C. (2023, January). *Federated learning: An introduction* (tech. rep.). The Open Data Institute (ODI). Retrieved January 14, 2024, from https://theodi.org/insights/reports/federated-learning-an-introduction-report/

Siddarth, D. (n.d.). Four Approaches to Democratizing AI. Retrieved January 2, 2024, from https://cip.org/research/democratizing-ai

Singh, A., & Hardinges, J. (2022, March). *Measuring the impact of data institutions* (tech. rep.). The Open Data Institute (ODI). Retrieved January 14, 2024, from https://theodi.org/insights/reports/measuring-the-impact-of-data-institutions-report/

Singh, P. J. (2018). Digital Industrialisation in Developing Countries: A Review of the Business & Policy Landscape (UNCTAD Report).

Singh, P. J. (2019, November). Data and Digital Intelligence Commons (Making a Case for their Community Ownership). https://doi.org/10.2139/ssrn.3873169

Singh, R. (2021, January). Mapping AI in the Global South. Retrieved April 15, 2024, from https://medium.com/datasociety-points/ai-in-the-global-south-sites-and-vocabularies-e3b67d631508

Smith, M., & Neupane, S. (2018, April). *Artificial intelligence and human development : Toward a research agenda* (tech. rep.). International Development Research Centre Canada. Retrieved February 24, 2024, from http://hdl.handle.net/10625/56949

Snaith, B. (2023, December). What do we mean by 'without data, there is no AI'? Retrieved January 19, 2024, from https://theodi.org/news-and-events/blog/what-do-we-mean-by-without-data-there-is-no-ai/

Solaiman, I. (2023, February). The Gradient of Generative AI Release: Methods and Considerations [arXiv:2302.04844 [cs]]. Retrieved May 25, 2024, from http://arxiv.org/abs/2302.04844

Soni, S. (2020, June). *Data Stewardship – A Taxonomy* (tech. rep.) (Section: Featured). The Data Economy Lab. Retrieved January 14, 2024, from https://thedataeconomylab.com/2020/06/24/data-stewardship-a-taxonomy/

Spiekermann, K., Slavny, A., Axelsen, D. V., & Lawford-Smith, H. (2021). Big Data Justice: A Case for Regulating the Global Information Commons [Publisher: The University of Chicago Press]. *The Journal of Politics*, *83*(2), 577–588. https://doi.org/10.1086/709862

Stiglich, L. (2023, December). *Understanding the social and economic value of sharing data* (tech. rep.). Retrieved January 19, 2024, from https://theodi.org/

insights/projects/understanding-the-social-and-economic-value-of-sharing-data/

Stinson, C. (2022). Algorithms are not neutral. *AI and Ethics*, *2*(4), 763–770. https://doi.org/10.1007/s43681-022-00136-w

Stürmer, M., Tiede, M. A., Nussbaumer, J. M., & Wäspi, F. (2023). On Digital Sustainability and Digital Public Goods [Medium: application/pdf Publisher: Technische Universität Berlin]. https://doi.org/10.24451/ARBOR.19678

Taraborelli, D. (2015, May). The sum of all human knowledge in the age of machines: A new research agenda for Wikimedia. Retrieved January 13, 2024, from https://www.slideshare.net/dartar/the-sum-of-all-human-knowledge-in-the-age-of-machines-a-new-research-agenda-for-wikimedia-icwsm-15

Tarkowski, A. (2023, October). Falcon 180B, open source AI and control over compute [Publication Title: Open Future]. Retrieved November 14, 2023, from https://openfuture.eu/blog/falcon-180b-open-source-ai-and-control-over-compute

Tarkowski, A., & Warso, Z. (2024, March). *Commons-based Data Set Governance for AI* (tech. rep.). Open Future. Retrieved March 24, 2024, from https://openfuture.eu/wp-content/uploads/2024/03/240325_Commons_Based_Data_Set_Governance_for_AI.pdf

Thereaux, O., & Keller, J. R. (2021, March). *The Economic Impact of Trust in Data Ecosystems* (tech. rep.). Frontier Economics — The Open Data Insitute. Retrieved January 19, 2024, from https://theodi.org/insights/reports/the-economic-impact-of-trust-in-data-ecosystems-frontier-economics-for-the-odi-report/

Thierer, A. (2018, August). The Pacing Problem, the Collingridge Dilemma & Technological Determinism. Retrieved January 2, 2024, from https://techliberation.com/2018/08/16/the-pacing-problem-the-collingridge-dilemma-technological-determinism/

Tian, Z., Cui, L., Liang, J., & Yu, S. (2022). A Comprehensive Survey on Poisoning Attacks and Countermeasures in Machine Learning. *ACM Computing Surveys*, *55*(8), 166:1–166:35. https://doi.org/10.1145/3551636

Tsukayama, H. (2020, October). Why Getting Paid for Your Data Is a Bad Deal. Retrieved January 13, 2024, from https://www.eff.org/deeplinks/2020/10/why-getting-paid-your-data-bad-deal

Ulnicane, I., Eke, D. O., Knight, W., Ogoh, G., & Stahl, B. C. (2021). Good governance as a response to discontents? Déjà vu, or lessons for AI from other emerging technologies [Publisher: SAGE Publications]. *Interdisciplinary Sci-*

ence Reviews, *46*(1-2), 71–93. https://doi.org/10.1080/03080188.2020.1840220

van der Schaar, M. (2023, July). Reality-Centric AI. Retrieved February 7, 2024, from https://slideslive.com/39006433/realitycentric-ai?ref=og-meta-tags

Verdegem, P. (2022). Dismantling AI capitalism: The commons as an alternative to the power concentration of Big Tech. *AI & SOCIETY*. https://doi.org/10.1007/s00146-022-01437-8

Verhulst, S., Chafetz, H., & Zahuranec, A. (2023, November). Interwoven Realms: Data Governance as the Bedrock for AI Governance. Retrieved January 22, 2024, from https://medium.com/data-policy/interwoven-realms-data-governance-as-the-bedrock-for-ai-governance-ffd56a6a4543

Viljoen, S. (2021). A Relational Theory of Data Governance. *Yale Law Journal*, *131*, 573. https://heinonline.org/HOL/Page?handle=hein.journals/ylr131&id=595&div=&collection=

Villalobos, P., Sevilla, J., Heim, L., Besiroglu, T., Hobbhahn, M., & Ho, A. (2022, October). Will we run out of data? An analysis of the limits of scaling datasets in Machine Learning [arXiv:2211.04325 [cs]]. https://doi.org/10.48550/arXiv.2211.04325

Wang, A., Liu, A., Zhang, R., Kleiman, A., Kim, L., Zhao, D., Shirai, I., Narayanan, A., & Russakovsky, O. (2022). REVISE: A Tool for Measuring and Mitigating Bias in Visual Datasets. *International Journal of Computer Vision*, *130*(7), 1790–1810. https://doi.org/10.1007/s11263-022-01625-5

Warso, Z., Keller, P., & Tarkowski, A. (2023). Exploring the Intersection of Openness and AI: Questions for Consideration and Collaborative Dialogue [Publisher: Open Future Foundation]. *Open Future*. Retrieved January 14, 2024, from https://openfuture.pubpub.org/pub/questions-on-ai-ml-and-openness/release/2

Whittlestone, J., & Ovadya, A. (2020, January). The tension between openness and prudence in AI research [arXiv:1910.01170 [cs]]. Retrieved January 2, 2024, from http://arxiv.org/abs/1910.01170

Wilson, B., Hoffman, J., & Morgenstern, J. (2019, February). Predictive Inequity in Object Detection [arXiv:1902.11097 [cs, stat]]. Retrieved April 14, 2024, from http://arxiv.org/abs/1902.11097

Wolfe, R., & Caliskan, A. (2022). American == White in Multimodal Language-and-Image AI. *Proceedings of the 2022 AAAI/ACM Conference on AI, Ethics, and Society*, 800–812. https://doi.org/10.1145/3514094.3534136

Yakowitz, J. (2011). Tragedy of the Data Commons. *Harvard Journal of Law & Technology*, *25*(1).

Zakardis, G. (2020, October). How data trusts can democratize the AI economy and accelerate innovation. Retrieved February 27, 2024, from https://www.atlanticcouncil.org/blogs/geotech-cues/how-data-trusts-can-accelerate-innovation/

Zarkadakis, G. (2020). "Data Trusts" Could Be the Key to Better AI [Section: Corporate social responsibility]. *Harvard Business Review*. Retrieved February 27, 2024, from https://hbr.org/2020/11/data-trusts-could-be-the-key-to-better-ai

Zhao, J., Wang, T., Yatskar, M., Ordonez, V., & Chang, K.-W. (2018, June). Gender Bias in Coreference Resolution: Evaluation and Debiasing Methods. In M. Walker, H. Ji, & A. Stent (Eds.), *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 2 (Short Papers)* (pp. 15–20). Association for Computational Linguistics. https://doi.org/10.18653/v1/N18-2003

Zuboff, S. (2019, January). *The Age of Surveillance Capitalism: The Fight for a Human Future at the New Frontier of Power: Barack Obama's Books of 2019* [Google-Books-ID: W7ZEDgAAQBAJ]. Profile Books.

Zygmuntowski, J. J., Zoboli, L., & Nemitz, P. F. (2021). Embedding European values in data governance: A case for public data commons. *Internet Policy Review*, *10*(3). https://doi.org/10.14763/2021.3.1572

Statement of Authorship

I hereby confirm and certify that this master thesis is my own work. All ideas and language of others are acknowledged in the text. All references and verbatim extracts are properly quoted and all other sources of information are specifically and clearly designated. I confirm that the digital copy of the master thesis that I submitted on 06/06/2024 is identical to the printed version I submitted to the Examination Office on 07/06/2024, 2024.

Date: **06/06/2024**

Name: **Henry Baker**

Student ID: **228755**

Signature: